



Speech based detection of Alzheimer's disease: a survey of AI techniques, datasets and challenges

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Abstract

Alzheimer's disease (AD) is a growing global concern, exacerbated by an aging population and the high costs associated with traditional detection methods. Recent research has identified speech data as valuable clinical information for AD detection, given its association with the progressive degeneration of brain cells and subsequent impacts on memory, cognition, and language abilities. The ongoing demographic shift toward an aging global population underscores the critical need for affordable and easily available methods for early AD detection and intervention. To address this major challenge, substantial research has recently focused on investigating speech data, aiming to develop efficient and affordable diagnostic tools that align with the demands of our aging society. This paper presents an in-depth review of studies from 2018–2023 utilizing speech for AD detection. Following the PRISMA protocol and a two-stage selection process, we identified 85 publications for analysis. In contrast to previous literature reviews, this paper places a strong emphasis on conducting a rigorous comparative analysis of various Artificial Intelligence (AI) based techniques, categorizing them meticulously based on underlying algorithms. We perform an exhaustive evaluation of research papers leveraging common benchmark datasets, specifically ADReSS and ADReSSo, to assess their performance. In contrast to previous literature reviews, this work makes a significant contribution by overcoming the limitations posed by the absence of standardized tasks and commonly accepted benchmark datasets for comparing different studies. The analysis reveals the dominance of deep learning models, particularly those leveraging pre-trained models like BERT, in AD detection. The integration of acoustic and linguistic features often achieves accuracies above 85%. Despite these advancements, challenges persist in data scarcity, standardization, privacy, and model interpretability. Future directions include improving multilingual recognition, exploring emerging multimodal approaches, and enhancing ASR systems for AD patients. By identifying these key challenges and suggesting future research directions, our review serves as a valuable resource for advancing AD detection techniques and their practical implementation.

Keywords Dementia · Alzheimer's disease · Classification · Regression · Speech data · Machine learning · Deep learning

1 Introduction

While dementia is an overarching term that refers to a range of symptoms affecting cognitive abilities, Alzheimer's disease (AD) is a specific type of dementia characterized by changes in the brain, including the accumulation of amyloid β plaques and neurofibrillary tangles (Scheltens et al. 2021). When this happens, neurons and neural connections are damaged, impacting the person's capacity to remember, think, and eventually create obvious functional impacts on daily life (Scheltens et al. 2021). There are other types of dementia, each with its own distinct causes and characteristics (Alzheimer's Association 2023). In this survey, we focused on AD which is also one of the most common types of dementia today, accounting for more than 60–80% of cases (Alzheimer's Association 2023). Driven primarily by aging populations, the number of people living with dementia globally was estimated to be 55 million in 2019, and this number is expected to rise to 139 million in 2050, according to the most recent report from the World Health Organization (WHO) (2021). Dementia is associated not only with a substantial personal cost to individuals and their families but also with enormous healthcare burdens and costs to the whole society. The total cost of health care, long-term care, and hospital services for people aged 65 and older with dementia is projected to be \$345 billion in the United States in 2023 (Alzheimer's Association 2023). However, the situation in developing countries is even worse, based on the World Alzheimer Report (2022), which estimates that 75% of people with dementia are not diagnosed globally, with that rate believed to rise as high as 90% in some lower- and middle-income countries. Furthermore, the progression of Alzheimer's disease, from initial brain changes that are unnoticeable by the affected individual to changes that cause memory problems and eventually lead to physical disability, is referred to as the AD continuum in Fig. 1.

Pharmacological interventions for AD are limited to managing symptomatic manifestations, without providing curative or preventive effects for the underlying disease process (Breijyeh and Karaman 2020). Although memory loss and cognitive decline are among the primary indicators of AD, it is worth noting that language impairment is another preva-

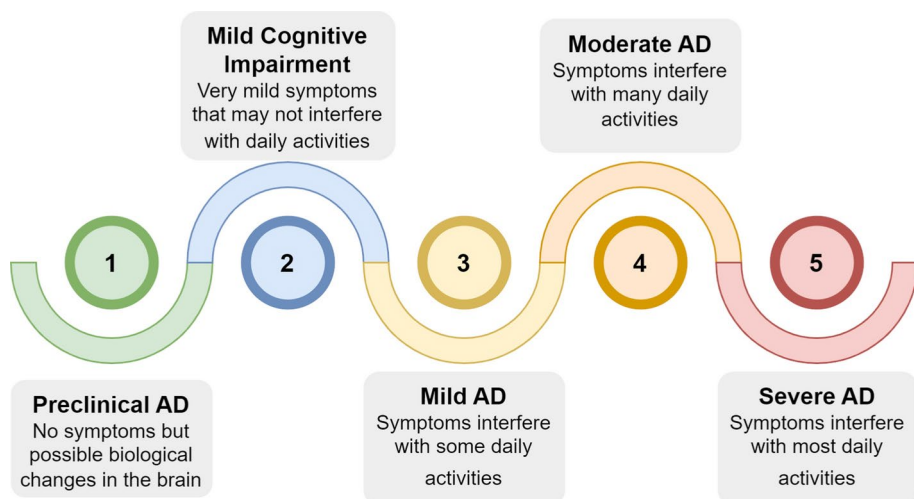


Fig. 1 Alzheimer's disease continuum (Alzheimer's Association 2023)

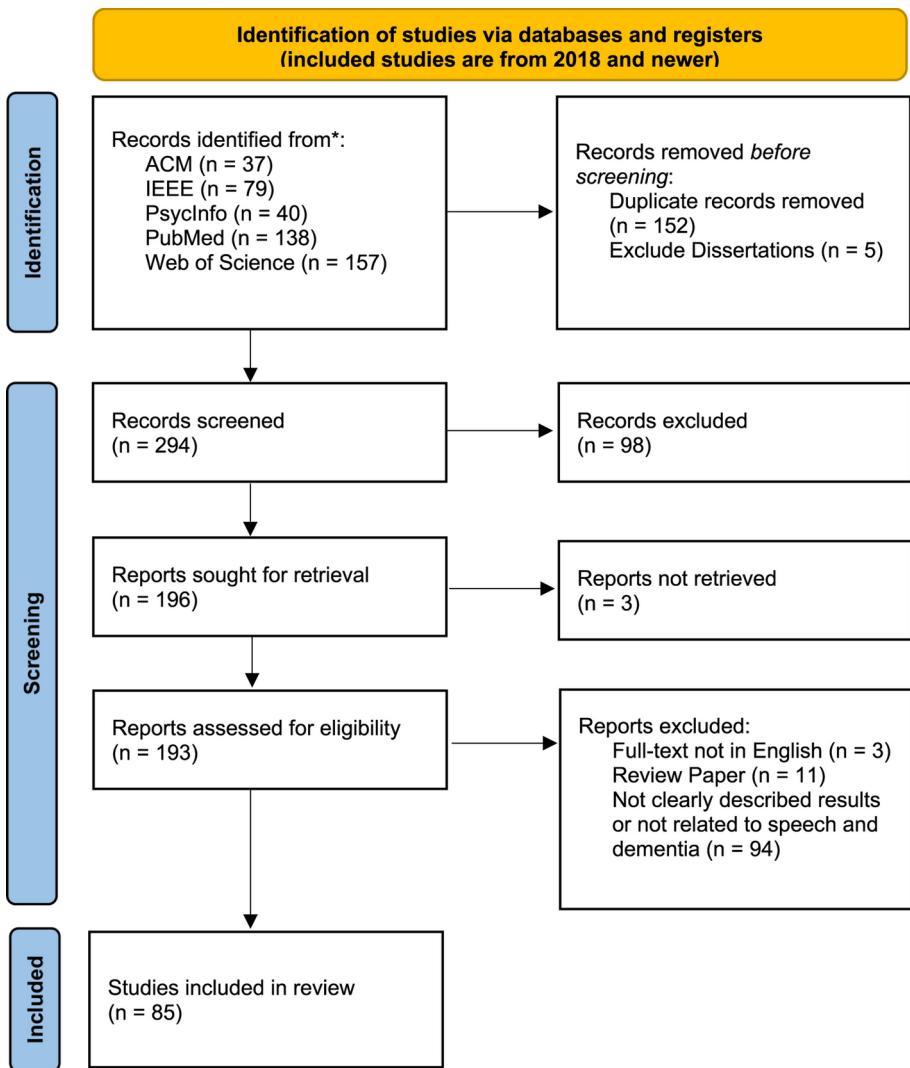


Fig. 2 Article selection process for literature review

lent symptom that can assist in diagnosing the condition and assessing its severity. This is because speech can provide insights into an individual's cognitive state and reveal additional aspects of brain damage. Several advantages of using speech as a biomarker for AD have been highlighted by prior research (de la Fuente Garcia et al. 2020; Hajjar et al. 2023), including the following:

1. Speech can be recorded and monitored over time, helping to reduce stress or anxiety in the clinical setting.
2. Noninvasive and cost-effective compared to other assessments, such as brain magnetic resonance imaging (MRI).

3. The technologies for speech analysis have significantly improved over the past ten years, supported by advancements in deep learning.
4. Speech impairments occur at various phases of the disease, demonstrating the significance of speech in distinguishing the stage of AD and forecasting the progression of the disease.

Currently, AD is diagnosed by a specialist physician who performs a series of clinical assessments: obtaining a personal and informant history of cognitive symptoms, a physical examination, cognitive tests on paper, blood tests to rule out other conditions that can mimic AD (such as low vitamin B12 levels), and brain scans to look for localized brain atrophy (Li et al. 2022b). The entire diagnostic procedure is time-consuming, expensive, invasive, and somewhat subjective because it is mainly based on the interpretation of the clinician. Therefore, it is necessary to have objective and cost-effective biomarkers to detect AD at an early stage.

Recent advancements in AI, particularly in the domains of speech recognition and Natural Language Processing (NLP), present promising prospects for diagnosing AD (Chen et al. 2023b; Dong et al. 2024). Extensive research surveys have been conducted in this area. Voleti et al. (2019) provided a comprehensive review of existing speech and language features to evaluate cognitive and thought disorders. Thaler and Gewald (2021) focused on identifying acoustic, linguistic, and demographic characteristics of AD themselves instead of comparing any machine learning algorithms. Boletsis (2020) discussed intelligent virtual agents in interaction aspects of speech-based dementia screening. Javeed et al. (2023) focused on all types of modalities such as neuroimaging, clinical feature-based data, and voice data more broadly rather than trying to recognize AD through speech or transcript data only. Although previous studies have proposed diverse signal processing and machine learning techniques to address this issue, the lack of balanced and standardized datasets for systematic comparisons of these different approaches remains a significant challenge within the field. This issue has been highlighted in prior systematic reviews (de la Fuente Garcia et al. 2020).

However, in the past three years, notable advancements have been made in the field of AD research, particularly in addressing this inadequate standardization. It has been achieved through the introduction of benchmarks such as ADReSS (Alzheimer's Dementia Recognition through Spontaneous Speech) (Luz et al. 2020) and ADReSSo (Alzheimer's Dementia Recognition through Spontaneous Speech Only) (Luz et al. 2021). These benchmark challenges have played a pivotal role in overcoming the existing obstacles that impede the translation of research findings into clinical practice.

This study distinguishes itself from previous research efforts by providing a systematic overview of the current methodologies used in the recognition of AD based on different methods. Additionally, it aims to enhance the comparability of algorithms by ensuring uniform evaluation conditions and requirements, as exemplified by the ADReSS challenge (Luz et al. 2020). Therefore, this review encompasses three main objectives:

- (1) Summarizing and critically evaluating existing diagnostic approaches for AD that leverage speech data. These methods are categorized based on the various machine learning techniques they employ.

- (2) Tackling the current challenges present in this field by employing a consistent benchmark dataset for meaningful comparisons among all studies.
- (3) Delving into additional ongoing challenges and investigates potential directions for future research. The remainder of this paper is structured as follows. Section 2 elaborates on the methodology used to retrieve prior research articles. In Sect. 3, an overview of the public dataset and data type are presented. Section 4 presents an overview of the current methods based on various machine learning techniques. In Sect. 5, an overview of the benchmark comparison is presented. Section 6 presents a discussion of aggregated findings from various studies to provide a comprehensive view. In Sect. 7, the current challenges regarding dataset, privacy and interpretability are discussed. Section 8 presents the future research directions and finally the paper is concluded in Sect. 9.

2 Systematic literature review methodology

This section presents the methodology used to retrieve and select research articles for this systematic literature review on artificial intelligence methods in detecting AD through speech analysis.

2.1 Source of the study

Between July and August 2023, we searched the following five electronic search engine: ACM, IEEE, PsycINFO, PubMed and Web of Science. We also included relevant titles found through 'forward citation tracking' with Google Scholar, selection articles references, and research portal suggestions.

2.2 Search criteria

The search queries were tailored to each search engine. An example query for PubMed is as follows:

- (speech AND (dementia OR "cognitive decline" OR (cognit* AND impair*) OR Alzheimer) AND ("machine learning" OR "deep learning" OR "natural language processing" OR "signal processing"))
- Filters applied: 01/01/2018–31/07/2023

2.3 Study inclusion and exclusion criteria

We applied predefined inclusion and exclusion criteria to select relevant articles for our review.

Inclusion criteria:

- The articles describing artificial intelligence applications to aid in the detection of de-

mentia/Alzheimer's disease through speech.

- The articles clearly describe results of classifying normal cognition, mild cognitive impairment, and dementia/Alzheimer's disease, while providing specific performance metrics such as accuracy, precision, recall, F1-score, or other relevant indicators.
 - The articles were peer-reviewed.
 - The articles were written in English.
- Exclusion criteria:*
- The articles did not have clearly specified classification results.
 - The articles were review papers instead of original works.
 - The articles using transitional statistics in the analysis instead of machine learning methods.
 - The articles that did not relate to speech or dementia.

2.4 Selection process

The selection process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al. 2021), ensuring a rigorous and unbiased approach. The process was conducted in two phases:

1. Initial screening:

- Titles and abstracts were screened against the exclusion criteria using Zotero.
- Duplicate articles were removed using the duplicate detection feature in Zotero.

2. Full-text screening:

- Articles that could not be definitively included or excluded based on titles and abstracts were subjected to a full-text review. The final set of articles included in this review totaled 85. The study flow and different phases of article selection are illustrated using the PRISMA 2020 diagram (Fig. 2).

2.5 Data collection process

A standardized approach was utilized to extract data from the selected studies. This process aimed to capture comprehensive information regarding each study's objectives, dataset, methodologies, major findings, and contributions to speech-based detection of AD. For each study included in the review, we extracted the following key information:

- The primary objectives of the study.
- Detailed descriptions of the machine learning and deep learning techniques applied.
- Preprocessing and feature extraction methods are used in speech analysis.
- The dataset used in the study.

- Performance metrics such as accuracy or F1-score, were used in this study.
- Any challenges or limitations identified in the study, particularly those related to feature extraction and model performance.
- Proposed directions for future research and potential improvements in methodology.

3 Dataset

3.1 Public dataset

The quality of a dataset is important in the machine learning process, as it significantly influences model performance. However, compared to other modalities such as neuroimaging, publicly accessible speech data is relatively limited. Table 1 presents several available datasets utilized in this field. Section 5 will provide a comprehensive analysis of the most commonly used dataset benchmarking.

Table 1 Summary of commonly used speech and text datasets

Dataset	Population size/ samples size	Modalities	Task type	Language	Avail- ability
Pitt Corpus (Becker et al. 1994)	HC: 104, AD: 208/552	Audio, text	PD, VF, SR, SC	English	OA
ADReSS (Luz et al. 2020)	HC: 78, AD: 78/156	Audio, text	PD	English	OA
ADReSSo (Luz et al. 2021)	HC: 115, AD: 122/237	Audio	PD	English	OA
ADReSS-M (Luz et al. 2023)	HC: 143, AD: 148/291	Audio	PD	English, Greek	OA
TAUKADIAL (Garcia et al. 2024)	169/507	Audio	PD	English, Chinese	OA
VAS Corpus (Kurtz et al. 2023)	HC: 36, MCI: 35, AD: 30/–	Audio	VC	English	OA
Delaware corpus (Lanzi et al. 2023)	HC: 24, MCI: 46 (ongoing)/–	Audio, text	PD, SR	English	OA
Dem@Care (Karakostas et al. 2016)	–/32	Audio, video, sensors	PD, R, SR	Greek	CA
Audio datasets of Framingham heart study (Xue et al. 2021)	HC: 291, MCI: 309, AD: 223/1264	Audio	PD, VF, SR, MF	English	RA
Carolina conversations collections (Pope and Davis 2011)	HC: 125, AD: 125/600	Audio	C	English	CA
Dementia Blog Corpus (Masrani et al. 2017)	HC: 3, AD: 3/2805 posts	Text	Posts	English	OA
Multimodal Dementia Corpus (Gkoumas et al. 2024)	HC: 10, AD: 12/816 sessions	Audio, text, keyboard, pen	C	English	RA

HC health control, MCI mild cognitive impairment, AD Alzheimer's disease, PD picture description, VF verbal fluency, SR story recall, SC sentence construction, R repetition, MF math fluency, C conversation, OA open access, CA conditional access, RA restricted access

Table 2 Summary of machine learning methods used for AD detection (Part 1)

Category	Subcategory	Method	Study
Traditional learning	Supervised learning	Support vector machine (SVM)	Parsapoor et al. (2023), Mei et al. (2023), Shah et al. (2023), Chen et al. (2023b), Vats et al. (2022), Luz et al. (2020), Luz et al. (2021), MohamedShreif and Lawgali (2022), Liang et al. (2022), Horigome et al. (2022), Woszczyk et al. (2022), Deng et al. (2022), Ablimit et al. (2022), Vats et al. (2021), Valsaraj et al. (2021), Triapthi et al. (2021), Syed et al. (2021), Searle et al. (2020)
		Linear regression	Luz et al. (2020), Luz et al. (2021), Valsaraj et al. (2021)
		Logistic regression	Shah et al. (2023), Burke et al. (2023), Horigome et al. (2022), Triapthi et al. (2021), Roshanzamir et al. (2021)
		Decision tree (DT)	Parsapoor et al. (2023), Vats et al. (2022), Luz et al. (2020), Luz et al. (2021), Liang et al. (2022), Vats et al. (2021)
		Naive Bayes (NB)	Vats et al. (2022), Horigome et al. (2022), Triapthi et al. (2021)
	Unsupervised learning	Linear discriminant analysis (LDA)	Parsapoor et al. (2023), Luz et al. (2020), Luz et al. (2021), Ablimit et al. (2022), Vats et al. (2021)
		K-nearest Neighbours (KNN)	Parsapoor et al. (2023), Luz et al. (2020), Luz et al. (2021), Vats et al. (2021)
		Gaussian mixture models (GMM)	Ablimit et al. (2022)
		Convolution neural network (CNN)	Igarashi et al. (2023), Chatzianastasis et al. (2023), Qian et al. (2022), Pandey et al. (2021), Lin et al. (2022), Meghanani et al. (2021)
		Long-short term memory (LSTM)	Zheng et al. (2022), Nambiar et al. (2022), Bouazizi et al. (2022), Meerza et al. (2022), Meghanani et al. (2021)
Deep learning		Gated recurrent unit (GRU)	Nambiar et al. (2022)
		Multilayer Perceptron (MLP)	Yadav et al. (2022), Bertini et al. (2022a), Meerza et al. (2022)
		Autoencoders (AE)	Vetráb et al. (2022), Bertini et al. (2022a)
		Attention-based model	Wen et al. (2023), Zheng et al. (2022), Lin et al. (2022), Ilias et al. (2022)

Table 2 (continued)

Category	Subcategory	Method	Study
Transfer learning	Text-based models	BERT, GPT, XLNet	Priyadarshinee et al. (2023), Pan et al. (2021), Ilias et al. (2023), Chen et al. (2023b), Chatzianastasis et al. (2023), Zhu et al. (2021b), Vats et al. (2022), Pérez-Toro et al. (2022), Pandey et al. (2021), Nambiar et al. (2022), Matošević and Jović (2022), Li et al. (2022a), Ilias et al. (2022), Ilias and Askounis (2022b), Horigome et al. (2022), Woszczyk et al. (2022), Deng et al. (2022) Agbavor and Liang (2022), Zhu et al. (2021a), Vats et al. (2021), Valsaraj et al. (2021), Roshanzamir et al. (2021), Pappagari et al. (2021), Balagopalan et al. (2021), Pappagari et al. (2020)
	Audio-based models	Wav2Vec2, VGGish, Whisper, HuBERT, XLSR-53, Mockingjay, YAMNet	Priyadarshinee et al. (2023), Pan et al. (2023), Pan et al. (2021), Mei et al. (2023), Chen et al. (2023b), Chen et al. (2023a), Zhu et al. (2021b), Pérez-Toro et al. (2022), Pandey et al. (2021), Zhu et al. (2021a), Chlasta and Wolk (2021)

3.1.1 Pitt corpus

This dataset was collected as a part of a project named TalkBank, which has been done as a part of the Alzheimer Research Program at the University of Pittsburgh. The main corpus in the DementiaBank (DB) is the Pitt corpus, which was collected longitudinally, between 1983 and 1988, every year from approximately 200 patients with AD and 100 healthy controls (Becker et al. 1994). This is one of most used dataset which has been made available to the wider research community via DementiaBank.

3.1.2 ADReSS

The Alzheimer's Dementia Recognition through Spontaneous Speech (ADReSS) dataset was developed as a refined subset of the Pitt corpus. This dataset provides a balanced distribution across classes, age, and gender, addressing some of the limitations present in its predecessor. The ADReSS dataset comprises audio recordings and corresponding transcriptions from 156 participants: 78 individuals diagnosed with AD and 78 HC subjects (Luz et al. 2020). While the ADReSS dataset represents a significant improvement in data balance, it is important to note a potential limitation. Because AD participants tend to use fewer words in their descriptions of the Cookie Theft picture compared to HC participants. This disparity in text volume could potentially introduce bias in models that rely solely on transcript analysis. Therefore, researchers should exercise caution when developing models based exclusively on transcription data and consider incorporating acoustic features or other modalities to mitigate this potential bias.

3.1.3 ADReSSo

Alzheimer's Dementia Recognition through Spontaneous Speech only (ADReSSo) provides more challenging and improved spontaneous speech recording compared with last year's ADReSS, requiring the creation of models straight from speech, without manual

Table 3 Summary of machine learning methods used for AD detection (Part 2)

Category	Subcategory	Method	Study
Transfer learning (continued)	Image-based models	ResNet, vision transformer (ViT), MobileNet	Ilias et al. (2023), Nishikawa et al. (2022), Ilias and Askounis (2022b), Zhu et al. (2021a)
		Random forest (RF)	Parsapoor et al. (2023), Burke et al. (2023), Vats et al. (2022), Luz et al. (2020), Luz et al. (2021), Liang et al. (2022), Kumar et al. (2022), Vats et al. (2021), Valsaraj et al. (2021), Triapthi et al. (2021), Martinc et al. (2021)
	Ensemble learning	Tree bagger (TB)	Pan et al. (2023); Luz et al. (2021)
		AdaBoost	Shah et al. (2021)
		Gradient boost	Nishikawa et al. (2022)
		Extreme gradient boosting (XGBoost)	Mei et al. (2023), Qian et al. (2022), Horigome et al. (2022)
		Majority voting	Mei et al. (2023), Chen et al. (2023b), Qian et al. (2022), Woszczyk et al. (2022), Deng et al. (2022)
	Multimodal Learning	Feature/tensor fusion	Pandey et al. (2021), Ilias et al. (2022)
		Averaging/score fusion	Mei et al. (2023), Chen et al. (2023b), Pandey et al. (2021), Deng et al. (2022), Pappagari et al. (2021), Pappagari et al. (2020)
		Add/concatenation fusion	Zhu et al. (2021a)

transcription, although automatic transcription is also encouraged (Luz et al. 2021). The ADReSSo dataset consists of 237 audio samples, with 115 samples from healthy controls and 122 from individuals with Alzheimer's disease. The audio recordings are in English and are openly accessible to researchers. However, same with ADReSS dataset, this dataset also focuses solely on the picture description task, which may not capture the full spectrum of cognitive impairments associated with Alzheimer's disease.

3.1.4 ADReSS-M

The ADReSS-M dataset, designed for the Multilingual Alzheimer's Dementia Recognition through Spontaneous Speech challenge at ICASSP 2023, represents a significant advancement in multilingual speech-based Alzheimer's detection. This dataset consists of 291 spontaneous speech samples, divided into 245 samples for training (237 in English and 8 in Greek) and 46 samples for testing (all in Greek) (Luz et al. 2023). Key features of the ADReSS-M dataset include:

1. **Balanced demographics:** the dataset is carefully balanced for age and gender to mitigate potential biases in the analysis.
2. **Standard labeling:** each sample is labeled with cognitive status (Control or ProbableAD) and includes Mini-Mental State Examination (MMSE) scores, providing a standardized measure of cognitive function.
3. **Dual-task challenge:** the dataset supports two primary tasks: (a) Binary classification of speech samples as healthy or AD (b) Regression task to predict MMSE scores. The multilingual nature of ADReSS-M is particularly noteworthy, as it aims to identify acoustic features that generalize across languages for AD detection. This approach is crucial for developing robust, language-independent tools for cognitive health monitoring. However, limitations of this dataset is the relatively small number of Greek samples in the training set, which may challenge the development of truly language-agnostic models.

3.1.5 TAUADIAL

The 2024 TAUADIAL Challenge dataset represents another step forward to multilingual speech analysis in cognitive impairment detection. This dataset contains 507 speech samples (261 in Chinese and 246 in English), collected during cognitive-linguistic battery assessments in clinical settings. The total duration of these samples is 528 min, providing a substantial corpus for analysis. With a ratio of just over 3:1, the dataset provides sufficient training data while retaining a significant portion for testing. The dataset has also been made available to the wider research community via DementiaBank, facilitating collaborative research efforts (Garcia et al. 2024).

3.1.6 VAS corpus

The Voice-Assistant Systems (VAS) dataset offers a unique perspective on cognitive impairment detection through the lens of voice assistant interactions. This corpus includes 101 participants divided into three groups: 36 healthy controls, 35 with mild cognitive impairment, and 30 with dementia (Kurtz et al. 2023). Key features of the VAS Corpus include:

1. **Authentic interaction scenario:** participants engage with an Alexa Echo device, imitating real-life interaction rather than performing a picture description task in a clinical environment.
2. **Diverse command set:** each participant was given 30 popular Alexa commands plus four additional basic commands, totaling 34 commands per participant.

3. Inclusion of MCI group: unlike Some datasets that focus only on healthy controls and AD patients, the VAS Corpus includes an MCI group, allowing for the study of early-stage cognitive decline.
4. Open access: the dataset is also available to the research community via Dementia-Bank, promoting collaborative research efforts. However, the VAS Corpus has some limitations:
 1. Command-based interaction: the reliance on pre-defined voice commands might not capture the full extent of language impairment, especially in free speech.
 2. Advanced dementia limitations: as noted in the study, one dementia patient failed to produce commands due to advanced dementia, indicating that this approach might not be suitable for very severe cases.
 3. Potential bias: the use of a specific voice assistant (Alexa) might introduce biases related to familiarity with the technology. Despite these limitations, the VAS Corpus provides valuable insights into how individuals with varying levels of cognitive function interact with voice-based technologies, which could inform both clinical assessment tools and the design of accessible voice interfaces.

3.1.7 Delaware corpus

The Delaware corpus (Lanzi et al. 2023), as part of the expanded DementiaBank, is an ongoing project aimed at providing a comprehensive dataset for analyzing language patterns in older adults at risk for dementia due to possible Alzheimer's disease. Key features of the Delaware corpus include:

1. Diverse participant pool: currently comprising 70 participants, including individuals without cognitive impairments and those with MCI.
2. Comprehensive protocol: the corpus includes four distinct types of language tasks: (a) Picture Descriptions: Cookie Theft, Cat Rescue, "Going and Coming" (Norman Rockwell) (b) Story Narrative: Cinderella (c) Procedural Narrative: Peanut Butter and Jelly Sandwich (d) Personal Narrative: Hometown
3. Focus on early detection: the inclusion of MCI participants makes this corpus particularly valuable for studying early linguistic markers of cognitive decline. Limitations of the Delaware corpus may include:
 1. Limited sample size: with only 70 participants at present, the dataset might have insufficient statistical power for conducting machine learning analyses.
 2. Lack of severe AD cases: the focus on MCI and healthy controls means the corpus may not represent the full spectrum of Alzheimer's disease progression.

3.1.8 Dem@Care

The Dem@Care dataset offers a comprehensive, multi-modal approach to studying dementia in both controlled and naturalistic settings (Karakostas et al. 2016). Key features of Dem@Care include:

1. Diverse data collection settings: data gathered from both lab experiments at the Greek Alzheimer's Association and participants' homes, providing a range of controlled and real-world contexts.
2. Multi-modal data: the dataset encompasses video and audio recordings, physiological sensor data, sleep sensor data, motion sensor data, and plug sensor data.
3. Controlled access: available to the research community under specific terms of use, ensuring ethical use of sensitive data. A potential limitation is cultural specificity, as the data gathered mainly in Greece might not be broadly applicable to other cultural environments.

3.1.9 Audio datasets of Framingham heart study

This dataset represents a significant resource for studying cognitive decline through speech analysis, leveraging the renowned Framingham Heart Study (FHS) (Xue et al. 2021). Key features include:

1. Large sample size: 1264 digital voice recordings, providing substantial statistical power.
2. Diverse cognitive status: includes 483 healthy controls, 451 with mild cognitive impairment, and 330 with dementia.
3. Longitudinal design: as part of the FHS, this dataset potentially allows for tracking cognitive changes over time.
4. Naturalistic recordings: average duration of 73 min per recording, capturing extended interactions between participants and clinicians.
5. Multiple cognitive tasks: includes recordings of various neuropsychological examinations. One significant limitation is the access restrictions. Local ethical committee regulations prevent this dataset from being publicly shared. It might be accessible to researchers upon request, thus restricting its use within the broader research community.

3.1.10 Carolina conversations collections

The Carolina Conversations Collection (CCC) (Pope and Davis 2011) is a comprehensive, password-protected digital archive of transcribed audio and video recordings focusing on health-related conversations with older adults. This unique corpus consists of two main cohorts:

1. A set of over 200 consented conversations with 125 older men and women from diverse ethnic backgrounds, each with at least one of 12 chronic medical conditions. These participants are recorded twice a year.
2. A longitudinal set of over 400 naturally occurring conversations with 125 persons with Alzheimer's disease, recorded at least twice annually. Key features of the CCC include:
 1. Diverse participant pool: the collection includes speakers from various racial, ethnic, and linguistic groups in North and South Carolina.
 2. Privacy measures: aliases are used for all participants, and a strict protocol is followed for de-identification of personal information.
 3. Accessibility: the corpus is housed at the Medical University of South Carolina (MUSC) Library and is accessible online to approved researchers under a restricted data-use agreement. Limitations of the CCC include:
 1. Limited geographic scope: the corpus is focused on speakers from North and South Carolina, potentially limiting its generalizability to other regions or countries.
 2. Potential selection bias: participants are volunteers, which may not represent the full spectrum of older adults or those with specific health conditions.
 3. Lack of standardized tasks: while valuable for naturalistic language study, the absence of fully standardized tasks across all interviews may make certain comparative analyses challenging.

3.1.11 Dementia blog corpus

The Dementia Blog Corpus (Masrani et al. 2017) offers a unique perspective on the written language of individuals with dementia and their caregivers, captured through the medium of public blogs. Key features includes:

1. Naturalistic data: 2805 blog posts provide genuine, unprompted written language samples.
2. Longitudinal aspect: blog posts over time potentially capture progression of language changes.
3. Varied dementia types: includes Alzheimer's Disease and Dementia with Lewy Bodies, allowing for comparison between different types of dementia. Limitations of the Dementia Blog Corpus include:
 1. Small sample size: only 6 blogs (3 with dementia, 3 controls) limit generalizability.
 2. Selection bias: bloggers may not be representative of the broader dementia population, as they are likely to be higher functioning and more technologically savvy.

3. Potential for edited content: blog posts may be edited or curated, potentially masking some language deficits.
4. Data loss: some blogs have become inaccessible or changed due to website updates, and the author only provided a web script rather than structured text data.

3.1.12 Multimodal dementia corpus

This longitudinal, multi-modal corpus represents a comprehensive approach to monitoring dementia through various forms of language production. Some key features include (Gkoumas et al. 2024):

1. Longitudinal design: data collected over time, allowing for tracking of disease progression.
 2. Multi-modal approach: incorporates speech (490 sessions), typed text (271 sessions), and hand-written text (104 sessions).
 3. Balanced groups: includes 12 participants with dementia and 10 healthy controls.
 4. Extensive data: 816 sessions in total, including 101:26 h of speech data.
 5. Detailed transcriptions: a subset of speech data (51 sessions from 8 participants) has been manually transcribed by professionals.
 6. Natural language use: uses reminiscence material in a non-clinical setting, aiming to capture more naturalistic language use. However, this Multimodal Dementia Corpus still has several limitations:
1. Small cohort size: with only 22 participants, generalizability may be limited.
 2. Potential task bias: while aiming for natural language use, the reminiscence tasks may still not fully reflect daily conversation and writing.
 3. Access restrictions: raw data is not publicly available, though it may be accessed through an NDA agreement.

3.2 Data type

Speech data in the context of Alzheimer's disease and cognitive impairment research can be categorized into several types based on different language tasks, as illustrated in Table 1. The first category is spontaneous speech, with the Cookie Theft picture description task from the Boston Diagnostic Aphasia Examination (Goodglass et al. 2001) being the most recognized task within this group. This task aims to evaluate the semantic knowledge and narrative abilities of subjects by asking them to describe a complex image. The Pitt Corpus, ADReSS, ADReSSo, and ADReSS-M datasets all incorporate this type of task. Another form of spontaneous speech is free conversation, as used in the Carolina Conversations Collections. Horigome et al. (2022) argued that free conversation without specific task requirements might be more suitable for clinical settings due to its naturalistic nature. This approach allows for the analysis of language use in a more relaxed, everyday context. The

second category is verbal fluency tasks, which include both phonemic and semantic fluency assessments. Phonemic fluency tasks involve generating words that start with a specific letter (e.g., “F” or “S”) within a limited time frame. Semantic fluency tasks require participants to produce words belonging to a particular category (e.g., “animals” or “fruits”). The Pitt Corpus and the Audio Datasets of the Framingham Heart Study incorporate verbal fluency tasks. The final category is comprised of various specialized tasks, including story recall and voice assistant commands, as illustrated by the Delaware and VAS Corpora, respectively. This study primarily focuses on spontaneous speech data, particularly in the context of the picture description task. This task has been widely employed in AD detection research due to its ability to collect complex language data in a standardized setting. Datasets such as ADReSS and ADReSSo have been specifically designed around this task, providing rich resources for developing and testing AD detection models based on spontaneous speech analysis.

4 Machine learning based methods for AD detection

Several machine learning methods are currently being employed for AD detection using speech-based data. These methods can be broadly categorized into traditional learning, deep learning, transfer learning, ensemble learning, and multimodal learning, as illustrated in Tables 2 and 3. In some studies (Luz et al. 2020, 2021; Vats et al. 2021; Triapthi et al. 2021), machine learning classifiers such as Support Vector Machines, Decision Trees, Naive Bayes, Random Forest, Multi Layer Perceptron, and 1-Nearest Neighbors have been commonly employed for comparative analyses subsequent to feature extraction from both voice recordings and textual transcripts. Therefore, these studies are not individually mentioned in each category of the machine learning method given below. It is essential to note that in many research studies, authors often employ a combination of these methods to construct a more comprehensive hybrid model.

Each of these categories encompasses a range of specific techniques, each with its own strengths and applications in AD detection. Traditional models, including Support Vector Machines and Logistic Regression, demonstrate efficiency when applied to well-crafted features extracted from both voice recordings and textual transcripts. Deep learning models, including Convolutional Neural Networks, have shown effectiveness in acoustic feature extraction from audio data. Transfer learning approaches leverage pre-trained language models like BERT to obtain linguistic features from transcripts. Ensemble learning methods, such as those based on voting and stacking, combine multiple models to enhance performance and produce more robust results. Multimodal learning techniques integrate various data types, allowing machine learning models to leverage complementary information from linguistic patterns, acoustic features, and visual representations of speech signals. In the following subsections, we will examine the various methods within each of these categories in detail.

4.1 Traditional learning

Traditional machine learning models, often referred as classical or conventional machine learning models, excel in scenarios with well-defined features in relatively small to medium-

sized datasets. They are more interpretable compared with other deep learning methods, making it easier to understand and explain their decision-making processes, which is crucial in healthcare domains where transparency and accountability are essential. However, unlike deep learning techniques, these conventional models may require the generation of domain-specific features or deep embedding features from other methods before being applied to each task.

Support vector machines (SVM): SVMs, introduced by Cortes and Vapnik (1995), are supervised learning models used for classification and regression tasks. SVMs find an optimal hyperplane that maximizes the margin between different classes. Shah et al. (2023) investigated language-agnostic speech features such as word-level duration, pause rate, and speech intelligibility. They used Principal Component Analysis (PCA) for dimensionality reduction and logistic regression for classification, achieving 69% accuracy. For the MMSE score prediction, they employed Support Vector Regression (SVR), achieving a Root Mean Squared Error (RMSE) of 4.8 on the ADReSS-M dataset. Notably, the study did not directly use transcripts or translations but found word-level features from the Whisper model highly valuable. Chen et al. (2023b) presented a cross-lingual AD detection method using features extracted from openSmile, XLSR-53, and RoBERTa. They applied SVM with a connected layer, achieving 69.6% accuracy and an RMSE of 4.788. This work demonstrated SVM's applicability in cross-lingual settings. MohamedShreif and Lawgali (2022) utilized linguistic features, including silence rate, to detect AD using an SVM classifier, achieving 88% accuracy on the DementiaBank dataset. This study showcased SVM's high accuracy with linguistic features, although it lacked detailed discussion on feature engineering. Syed et al. (2021) developed a multimodal AD detection system combining handcrafted and deep embeddings for acoustic and linguistic data. They achieved 91.67% accuracy on the ADReSS dataset using logistic regression, SVM, and majority voting. They found out that deep embeddings, especially from environmental sounds and music, outperformed handcrafted features in both audio and text modalities, highlighting their efficacy in speech-related applications. Shah et al. (2021) explored various algorithms and dimension reduction techniques on the ADReSS dataset, achieving 85.4% classification accuracy and an RMSE of 5.62. They combined logistic regression, random forest, SVM, and XGBoost in a weighted majority-vote ensemble, demonstrating SVM's effectiveness in ensemble methods. Searle et al. (2020) analyzed textual transcripts from the ADReSS dataset to classify AD. They employed various models, including SVM and Transformer-based models. The top-performing models were a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer input into an SVM model and a pre-trained DistilBERT model used as an embedding layer in simple linear models. These models achieved classification scores of 0.82 and an RMSE of 4.58, effectively capturing keyword frequency correlations despite performance variations across different datasets.

The strengths of SVM in AD detection include its effective handling of high-dimensional data and its suitability for both regression and classification tasks, while its weaknesses involve the need for careful feature extraction, challenges with large datasets, and sensitivity to noisy data and outliers.

Logistic regression (LR): LR determines the optimal coefficients for the linear combination of features that maximize the likelihood of the observed data, typically using optimization techniques such as gradient descent. While effective for linearly separable data, LR may struggle with complex, nonlinear relationships. Shah et al. (2023) applied logistic regres-

sion with L2 regularization to the top 10 principal components derived from meta-features, word-level duration, pause rate, and speech intelligibility. This approach achieved an accuracy of 69.57% on the Greek samples in ADReSS-M dataset. Burke et al. (2023) utilized logistic regression on Ostrand and Gunstad's lexical-semantic features, alongside newly developed lexical-semantic features, achieving a classification accuracy of 78.4% on the Pitt Corpus dataset. Roshanzamir et al. (2021) proposed transformer-based deep neural network language models trained on the Pitt corpus dataset. They explored various pretrained embedder layers (BERT, XLNet, XLM, GloVe) and classifier layers (logistic regression, neural networks, LSTM, CNN). Notably, using sentence-level BERT-Large with a logistic regression classifier resulted in accuracy and F1 scores of 88.08 and 87.23%, respectively.

Decision tree (DT): It is a versatile and interpretable machine learning approach used for both classification and regression tasks. It partitions input data into subsets based on feature values, forming a tree-like structure where each internal node represents a decision based on a feature, and each leaf node represents the prediction or outcome. Li and Huang (2021) proposed a task-oriented feature representation method for classifying AD based on spontaneous speech. The method incorporates classification task information into the feature representation learning process, improving classification accuracy compared to content-independent models that only use acoustic features. Experimental results on the ADReSS dataset demonstrated that DT model, utilizing the 1dCNN-triplet feature, surpassed linguistic features and achieved an accuracy of 83.3%. Parsapoor et al. (2023) conducted an empirical investigation to evaluate the impact of various language tasks, recording media, and modalities on machine learning classifiers for dementia assessment, including Decision Trees, Extra Trees, k-Nearest Neighbors, Support Vector Machines, Logistic Regression, and Random Forests. Their study revealed that classifiers trained on the picture description language task exhibited superior performance compared to those trained on the story recall language task. Additionally, phone-based recordings yielded more favorable outcomes compared to web-based recordings.

K-nearest neighbors (KNN): It is a simple and intuitive machine-learning algorithm that is used for both classification and regression tasks. KNN makes predictions based on the similarity between new data points and existing data points in a labeled dataset. Vats et al. (2021) focused on AD detection using BERT and eight different acoustic features plus a set of ComParE features (Eyben et al. 2013) (includes energy, spectral, MFCC, and voicing related low-level descriptors) to identify the attributes affected by AD in human speech production. Authors demonstrated the complementary nature of acoustic features to linguistic features in AD classification, showing an improvement of 6.1% classification accuracy when combining the acoustic model, the BERT model, and score fusion with the KNN algorithm. The final accuracy on the ADReSS test dataset is 85.2%.

Gaussian mixture models (GMMs): It is a probabilistic model used in machine learning for representing and modeling data that can be best described as a combination of multiple Gaussian (normal) distributions. GMMs are particularly useful when dealing with complex data that cannot be easily represented by a single Gaussian distribution. Ablimit et al. (2022) used two distinct corpora, the Interdisciplinary Longitudinal Study on Adult Development and Aging (ILSE) and the ADReSS challenge datasets. The speech and language features extracted from these two corpora were analyzed. For the ADReSS dataset, a GMM classifier achieved 66.7% accuracy using only ECAPA-TDNN acoustic features. ECAPA-TDNN embeddings are deep neural network-based embeddings that achieve state-of-the-art per-

formance in speaker recognition, were extracted by author using the model made available by SpeechBrain. In contrast, for the ILSE dataset, the GMM classifier achieved 83.8% accuracy using Part-of-Speech (PoS) linguistic features. This disparity highlights that the best classifiers and feature sets for one dataset do not necessarily match the best for another dataset, raising questions about the transferability of methods to new domains, languages, tasks, and recording conditions.

4.2 Deep learning

Due to the ability to model complex patterns and relationships in data, deep learning methods have been used in various fields. These methods are characterized by the use of artificial neural networks, particularly deep neural networks with multiple layers to extract and learn hierarchical features from raw audio and transcript data directly.

Convolution neural network (CNN): CNNs are specialized neural networks designed for processing grid-like data, such as images. In the context of AD detection, many authors used convolution layers to automatically learn and detect hierarchical features from input Mel Frequency Cepstral Coefficients (MFCC) or Mel-spectrogram data which is the representation of the short-term power spectrum of a audio signal. Igarashi et al. (2023) used CNN to classify 29 participants (7 males and 22 females) aged 72–91 years with moderate or mild dementia and mild cognitive impairment. They found out that Mel-spectrogram generally outperformed Mel Frequency Cepstral Coefficients in terms of accuracy, precision, recall, and F1-score in all classification tasks. Chatzianastasis et al. (2023) used Neural Architecture Search (NAS) method—DARTS to automatically find a high performing CNN architecture based on log-Mel spectrograms, then they exploit several fusion methods, including Multimodal Factorized Bilinear Pooling and Tucker Decomposition to capture inter-modal interactions. The overall performance on the ADReSS dataset achieved 92.08% in accuracy. Lin et al. (2022) used a combination of Transformer and CNN architecture for feature extraction from Mel-spectrogram. On their own 120 participants dataset with a balanced distribution of individuals with normal aging, mild cognitive impairment (MCI), and AD patients, their trials obtained an accuracy of 91 and 79% for identifying groups of AD and MCI from those with normal ageing, respectively. Chlasta and Wolk (2021) focused on using computer-based automated screening of dementia through spontaneous speech only. Two feature extraction methods were proposed: VGGish, a deep, pre-trained Tensorflow model, and DemCNN, a custom raw waveform-based convolutional neural network model. The VGGish model achieved an accuracy of 59.1%, while the DemCNN model achieved an accuracy of 63.6% on the ADReSS dataset.

Long-short term memory (LSTM): It is widely used in AD detection tasks due to its effectiveness in modeling sequential and time series data such as audio transcript. Zheng et al. (2022) made use of Howard and Ruder's pre-trained language mode—Averaged Stochastic Gradient Descent Weight-Dropped LSTM (AWD-LSTM) which trained using Wikipedia articles with about 103 million different words. They fine-tune this pre-trained language model by using the dementia data set in hand and adjust the language model for classification by using (ReLU) activation and softmax activation to substitute the last softmax layer in the original network. The overall accuracy on the Pitt Corpus dataset is 76.16%. Bouazizi et al. (2022) also utilized the AWD-LSTM method to detect dementia. However, they focused on context words as well as stop words with patterns of part-of-speech sequences as fea-

tures. Their experiments demonstrate the importance of both grammar and vocabulary in accurately identifying dementia and achieved 81.54% accuracy on the Pitt Corpus dataset. Rohanian et al. (2021) presented two multimodal fusion-based deep learning models for AD recognition using acoustic, lexical, disfluency, and speech pause features. The best model, BiLSTM with highway layers, achieved an accuracy of 84% and RSME error prediction of 4.26 on MMSE cognitive scores for ADReSSo Dataset, showing improvement over word-only models. The models effectively deal with noisy inputs from acoustic features and ASR hypotheses.

Gated recurrent unit (GRU): It is an enhanced interpretation of the traditional recurrent neural network. The GRU is just like a long short-term memory; however, it doesn't have an output gate, so it has fewer parameters. Nambiar et al. (2022) extracted vector representations using RoBERTa from Pitt Corpus dataset and this word embedding feed into GRU for Dementia detection. The overall accuracy is 80% slightly less than the BERT + BiLSTM model which achieved 81% accuracy in the experiment.

Autoencoders (AE): Autoencoders have a long history in machine learning, dating back long before deep networks. The basic idea is to encode the input into a compressed and meaningful representation and then decode it back such that the reconstructed input is as similar as possible to the original one (Bank et al. 2023). This compressed bottleneck layer can be used as features in a potential classification step. Vetráb et al. (2022) extracted Mel-scale spectrograms from raw waveforms and fed into autoencoder feature extractor models which used 2 recurrent layers, each one consisting of 128 GRU cells, and a bidirectional decoder. The autoencoder-based features are then used by a linear SVM for classification tasks and achieve an accuracy of around 72%. Bertini et al. (2022a) used the spectrogram of the audio signal as the feature for training the autoencoder network. They utilized a specific data augmentation approach called SpecAugment that avoids distorting the original samples, This proposed method achieves an accuracy of 93.30% and an F1 score of 88.50% when evaluated on the Pitt Corpus dataset. However, this paper did not explore the potential of using other types of features, such as acoustic, syntactic, and semantic features, which could enhance the classification performance and interpretability of the model.

4.3 Transfer learning

Transfer learning techniques leverage pre-trained models, often on large datasets, and fine-tune them for specific tasks. This approach has been particularly effective in scenarios where labeled data is scarce such as healthcare. Here is a review of some commonly used pre-trained models for AD detection.

BERT: The text transcriptions were used to generate feature embedding vectors from the pre-trained BERT (Bidirectional Encoder Representations from Transformers) model (Devlin et al. 2018). BERT is used for pretraining a language representation over a large amount of unlabeled textual data and employs an attention mechanism (an encoder that reads the text and a decoder that predicts), resulting in a deeper sense of language contextual relationships between words in a text. Valsaraj et al. (2021) focused on AD detection using acoustic and linguistic features, as well as a pre-trained BERT model. Three models were used for classification: Model 1 used acoustic features, Model 2 used linguistic features, and Model 3 used the auto-generated transcripts directly with a pre-trained BERT and TF-IDF. Their result in line with the previous studies where Deep Learning based linguistic models

outperform basic ML-based models and acoustic models. However, their result is not good enough compared with other BERT in the ADReSS benchmark dataset. Balagopalan et al. (2020) compared hand-crafted features and BERT-based models for AD detection, showing that fine-tuned BERT models perform well and outperform feature-based approaches. The fine-tuned BERT-based models achieve the highest AD detection accuracy of 83.3% on the ADReSS test set, outperforming the best feature-based classification model (SVM) with an accuracy of 81.3%. The authors also suggested that future work should combine BERT and hand-crafted features for improved performance.

RoBERTa: RoBERTa (a Robustly optimized BERT approach) (Liu et al. 2021a) uses dynamic masking (unlike BERT's static masking) which increases the data variability (by augmentation) and helps in yielding more robust features. Matošević and Jović (2022) used of speech transcripts from Pitt corpus only can lead to dementia detection rates above 90% for the RoBERTa model, which outperformed the baseline BERT model by 4 percent in accuracy. Priyadarshinee et al. (2023) systematically compare various modalities, granularities, and machine learning models for automated Alzheimer's Dementia detection. They tested sixteen features across audio and text modalities, analyzing the data at two levels: frame-level (granular or frame-by-frame descriptors) and file-level (across the entire interview recording). Their findings reveal that text-based classification utilizing RoBERTa outperformed audio-based classification, achieving an accuracy of 88.7% on the ADReSSo dataset.

DistilBERT: DistilBERT is a cheaper, smaller version of BERT (Sanh et al. 2019), which is 60% faster and 40% smaller, while retaining 97% of its performance. Pandey et al. (2021) utilized a pre-trained DistilBERT on the transcripts generated from speech signals using the Wav2vec2 model combined with CNN-based audio embeddings model which is employed to extract AD-related features from speakers' speech utterances. The BERT embedding and CNN-based audio embeddings are then used in score-level fusion to generate an average probability for AD classification. The accuracy on the test partition of the ADReSSo challenge dataset is 73.68%. Liu et al. (2022a) presented a transfer learning method for detecting AD based on DistilBERT embedding with a logistic regression classifier. This proposed model achieves an accuracy of 0.88, which is a considerable improvement over the baseline established by the challenge organizers.

GPT: Li et al. (2022a) proposed a novel method by using GPT-D, an artificially degraded version of GPT-2, which showed generalizability across different datasets, including ADReSS, DB, and CCC, with competitive performance compared to fine-tuning on BERT and DistilBERT models. Agbavor and Liang (2022) utilized GPT-3 text embeddings, which are vector representations of transcribed speech that capture the semantic meaning of the input. By combination with SVM, the GPT-3 text embeddings considerably outperform conventional acoustic feature-based approaches and even compete with fine-tuned models. Performance on ADReSSo achieved 80.3% in accuracy and 6.25 in RMSE. However the author did not explore other potential approaches or combinations of these features.

XLNet: It employs a Permutation Language Modeling (PLM) objective during pre-training. This approach involves shuffling sentence permutations and predicting the next word in each, exposing the model to diverse word orders and contexts (Yang et al. 2019). This comprehensive training strategy enhances XLNet's ability to understand intricate relationships within text. It used by (Priyadarshinee et al. 2023; Ilias and Askounis 2022a) to extract embedding from transcript to perform comparison evaluation.

Audio Models: Both HuBERT and Wav2Vec represent significant advancements in processing raw audio waveforms for speech recognition and analysis. Unlike previous language models such as BERT and XLNet, which focus on textual data, HuBERT adapts the Transformer architecture to handle the sequential nature of speech (Hsu et al. 2021), working directly with raw audio waveforms to extract pre-trained speech embeddings as acoustic features. This approach has been utilized by researchers such as Mei et al. (2023) and Chen et al. (2023a). Similarly, Wav2Vec employs self-supervised methods to transform raw audio into encoded representations for speech recognition systems (Baevski et al. 2020). Studies like Pandey et al. (2021) and Agbavor and Liang (2022) have used Wav2Vec2 to generate transcripts, demonstrating its efficacy in diverse applications.

In addition to the pretrained models discussed above, Zhu et al. (2021a) employ state-of-the-art deep transfer learning models from image, audio, speech, and language domains such as MobileNet, YAMNet, Mockingjay, and BERT. The authors compared the performance of different transfer learning models on ADReSS challenge dataset and show using text data significantly better performance than those using audio data. Multi-modal transfer learning introduces a slight improvement in accuracy which achieved 89.58% on ADReSS dataset. Yang et al. (2022) propose an augmented adversarial self-supervised learning approach for early-stage Alzheimer's speech detection. Unlike previous studies using public pre-trained models, authors develop their own pre-training method utilizing a large corpus of normal speech (AISHELL) alongside limited AD speech data. They employ contrastive predictive coding, adversarial training, and data augmentation techniques to capture AD-like patterns. The pre-trained model serves as a feature extractor for a downstream AD detection task. This approach significantly improves AD detection performance, especially for the AD class, achieving an F1-score of 83.73% on the 2021 NCMMSC Alzheimer's Disease Recognition Challenge, outperforming both baseline and wav2vec2.0-based models.

4.4 Ensemble learning

Ensemble learning is a machine learning technique that combines the predictions of multiple individual models to produce a more robust and accurate final prediction. The idea behind ensemble learning is that by aggregating the predictions of multiple models, the collective intelligence of the ensemble can often outperform any single model. Several papers have used ensemble methods for AD detection. Nishikawa et al. (2022) used LightGBM, a machine learning method for Gradient Boosting to calculate the Logloss between Mild Cognitive Impairment and Normal Controls through acoustic features. They also utilise the ViTb16, which is the Encoder part of the Vision Transformer (ViT) to analyse the Log-melspectrogram. Their experimental results show that the ensemble model of LightGBM and ViTb16 performs the best which has 90.7% accuracy. Qian et al. (2022) proposed a two-stage metric learning approach to detect subjects with AD, HC, and MCI. Because they think universal acoustic feature set as "silver bullet" to characterize all three groups does not exist. Each stage takes distinct acoustic features as input to a specific convolution neural network to get feature vector which then feed into XGboost classifier to group into AD or Non-AD and HC or MCI. Kumar et al. (2022) explored the use of standalone speech features to recognize dementia. They utilized speech samples from the Pitt corpus in DementiaBank and proposes a critical speech feature set of prosodic, voice quality, and cepstral features for the task. Random forest has been applied to these features and achieved

87.6% in accuracy. Hason and Krishnan (2022) proposed models which achieved accuracy rates of 82.2% for AD prediction and 71.5% for AD stages classification using a Random Forest classifier which is based on acoustic features extracted from ADReSSo challenge dataset. Nonetheless, the research has not explicitly demonstrated the methodology behind the extraction of these acoustic features. Martinc et al. (2021) employ a combination of acoustic, textual, and semantic bag-of-n-grams features for AD diagnosis based on spontaneous speech by using a random forest algorithm with 50 trees of maximum depth 5. This proposed approach achieves an accuracy of 93.75% on the ADReSS dataset. Alkenani et al. (2021) introduced a novel approach, fusion-based stacked generalization, aimed at enhancing the overall generalizability and robustness of machine learning models for AD diagnosis. This method leverages the strengths of multiple base learning algorithms to construct heterogeneous stacked fusion models. These fusion models are trained using two distinct datasets, one derived from written language (Dementia Blog Corpus) and the other from spoken language (DementiaBank). Notably, the stacked fusion models demonstrated remarkable performance, achieving area under the curve (AUC) values of 98.1 and 99.47% for the spoken and written-based datasets, respectively, while the hybrid model, trained on linked data, achieved optimal performance with an impressive AUC of 99.2%.

4.5 Multimodal learning

Multimodal Learning differs from ensemble learning in that it involves combining information from multiple modalities, such as text, images, and audio to make a decision. A prime example of Multimodal Learning is in Alzheimer's disease detection via speech analysis, where speech samples can be represented in multiple modalities, including text (transcripts), audio (raw recordings), and images (log-Mel spectrograms). By combining these modalities, Machine learning models can leverage the complementary information they provide, such as linguistic patterns, acoustic features, and visual representations of speech signals, to improve the accuracy of Alzheimer's detection. Normally, in Multimodal Learning, various fusion techniques are used to integrate the information from these three different modalities. For example, Score Fusion involves averaging prediction probabilities across different modalities for classifying sample into AD or Non-AD categories. Pappagari et al. (2021) enhanced this by using model output scores as inputs for a logistic regression classifier. They employed speech embedding techniques, such as x-vectors and prosody features, and used automatic speech recognition to obtain transcriptions for fine-tuning BERT models. Their results showed that score fusion from multiple acoustic and linguistic models provided the best detection results, with linguistic models generally outperforming acoustic models unless ASR errors impacted performance. Embedding Feature Fusion integrates audio and text features at a later stage, combining separately processed features from each modality for final classification (Pandey et al. 2021). Ilias et al. (2022) proposed Tensor Fusion, a method for detecting Alzheimer's disease from spontaneous speech. This approach uses deep neural networks to process audio, image, and text data, capturing inter-modal interactions through a tensor fusion layer. The method converts audio to multi-channel images, processes transcripts with BERT, and extracts acoustic features. By addressing limitations of earlier fusion methods, Tensor Fusion achieved 86.25% accuracy and 85.48% F1-score on the ADReSS dataset. Add Fusion, proposed by Zhu et al. (2021a), combines outputs from fully connected layers of two specific models: Text BERT and Speech BERT.

It selects the model with higher classification confidence and merges its features with the other model. This technique can enhance confidence in consistent classifications or favour the more confident model in cases of disagreement. Concatenation Fusion, also introduced by Zhu et al. (2021a), merges features from both Text BERT and Speech BERT models to create a hybrid representation. By combining outputs of fully connected layers from these two models, it leverages complementary information from text and speech modalities. Ilias and Askounis (2023) proposed novel multimodal approaches by utilizing BERT for transcript and DeiT for audio spectrograms image. By incorporating context-based self-attention, optimal transport domain adaptation, and label smoothing to improve both performance and model calibration, they achieved 91.25% accuracy and 91.06% F1-score based on ADReSS benchmark datasets.

5 Benchmarking

Benchmarking plays a crucial role in advancing AD recognition by providing standardized datasets, tasks, and evaluation metrics for comparative analysis. This section delves into prominent datasets commonly used in AD recognition research, as discussed in Sect. 3, with a particular focus on the ADReSS and ADReSSo datasets as benchmarks. These datasets enable researchers to consistently assess the performance of various machine learning and deep learning models.

5.1 Pitt corpus

The Pitt Corpus dataset is a widely used resource in AD recognition research. However, not all studies utilize the whole dataset due to the inherent issue of dataset imbalance within the entire corpus. This imbalance can lead models to focus on individual patient characteristics rather than general dementia-related features. A summary of research utilizing the Pitt Corpus dataset is provided in Table 4, which shows various machine learning and deep learning models along with their performance metrics.

5.2 ADReSS

The ADReSS dataset has been extensively used in AD recognition research. The ADReSS Challenge comprises two main tasks:

- **AD Classification Task:** developing a model to predict whether a speech session is AD or non-AD using speech data, manual transcript language data, or both.
- **MMSE Regression Task:** creating a model to infer the subject's Mini-Mental State Examination score based on speech and/or transcript language. The ADReSS Challenge baseline, established by Luz et al. (2020), provided an initial benchmark with 75% accuracy, enabling subsequent studies to demonstrate significant improvements. Table 5 summarizes various models with their performance metrics.

Table 4 Comparisons of research on Pitt corpus

Publication	ML/DL models	Accuracy	Precision	Recall	F1
Wen et al. (2023)	CNN-Attention-Feature Network	92.2%	93.5%	97.1%	95.5%
Begam et al. (2023)	Stacking ensemble learning (RF, SVM, NB, KNN)	78%	—	—	47%
Pan et al. (2023)	Tree bagger (TB)	79.04%	—	—	78.91%
Chen et al. (2023a)	Speechformer++	81.3%	—	—	80.8%
Burke et al. (2023)	Logistic regression and random forest	78.4%	—	85.7%	—
Zheng et al. (2022)	Attention-based models	81.54%	—	—	—
Nambiar et al. (2022)	BERT + BiLSTM	81%	79%	79%	79%
MohamedShreif and Lawgali (2022)	Support vector machine	88%	—	—	—
Matošević and Jović (2022)	RoBERTa	90.16%	92.81%	88.60%	90.28%
Kumar et al. (2022)	Random forest	87.6%	91.2%	82.2%	86.5%
Bouazizi et al. (2022)	Averaged stochastic gradient descent weight-dropped LSTM (AWDLSTM)	81.54%	87.34%	83.13%	85.19%
Liu et al. (2022b)	Transformer + Feature purification network	93.50%	94%	89%	91.19%
Triapthi et al. (2021)	BayesNet fuse with posterior two-class classifiers	—	85.9%	85.9%	85.7%
Roshanzamir et al. (2021)	$BERT_{Large}$ + Logistic regression (NLP)	88.08%	90.5%	84.34%	87.23%

5.3 ADReSSo

The ADReSSo dataset emphasizes creating models that work directly from speech without manual transcription, aligning closely with real-world applications. The ADReSSo Challenge includes the following tasks:

- **AD Classification Task:** developing a model to predict the label (AD or non-AD) for a speech session using either acoustic features or linguistic features extracted from automatically generated transcripts.
- **MMSE Regression Task:** creating a model to infer the subject's MMSE score based on speech data.
- **Cognitive Decline Inference Task:** predicting changes in cognitive status over time for a given speaker based on speech data collected at baseline. The ADReSSo Challenge baseline, established by Luz et al. (2021), set an initial benchmark with 78.87% accuracy. This benchmark has allowed subsequent studies to demonstrate significant improvements. Table 6 summarizes various models and their performance metrics.

The introduction of standardized benchmarks, particularly the ADReSS and ADReSSo datasets, has significantly impacted the field of AD recognition through speech analysis. These benchmarks have driven progress in several keyways:

Table 5 Comparisons of research on ADReSS dataset

Publication	ML/DL models	Accuracy	Precision	Recall	F1	RMSE
Luz et al. (2020) Baseline	LDA and decision tree	75%	83%	62%	71%	5.2
Ilias and Askounis (2023)	BERT + DeiT + Optimal transport + attention-based fusion (deep context) + label smoothing	91.25%	93.08%	89.17%	91.06%	–
Pan et al. (2023)	wav2vec2.0 + Tree bagger (TB)	82.64%	–	–	82.63%	–
Liu et al. (2023)	Tree bagger (TB)	70%	–	–	–	–
Ilias et al. (2023)	BERT + ViT + Crossmodal attention	88.33%	86.13%	91.67%	88.69%	–
Chatzianastasis et al. (2023)	DARTS + BERT + BLOCK	92.08%	94.09%	91.66%	91.94%	–
Aryal et al. (2022)	Stochastic gradient descent optimized model	–	–	–	–	3.9
Liu et al. (2022a)	DistilBert + Logistic regression with grid search	88%	88%	88%	87%	–
Ilias and Askounis (2022a)	BERT + Global average pooling layer + two dense layers	87.5%	87.19%	81.66%	86.73%	–
Ilias et al. (2022)	BERT + Swin transformer + Tensor fusion	86.25%	90.88%	80.83%	85.48%	–
Ilias and Askounis (2022b)	BERT + ViT + Gated self-attention	90.00%	90.87%	89.17%	89.94%	3.61
Woszczyk et al. (2022)	Audio spectrogram transformer(AST) + SVM + BERT + Majority vote	86%	–	–	–	–
Deng et al. (2022)	SVM + multi-head-self-attention (MHSA)-CNN + Majority vote	87.50%	88.57%	87.5%	87.41%	–
Meerza et al. (2022)	LSTM + MLP	89.1%	93.5%	92.3%	–	–
Ablimit et al. (2022)	GMM, LDA, SVM	77.1%	–	–	–	–
Liu et al. (2022b)	Transformer + Feature purification Network	78.6%	–	–	–	–
Haulcy and Glass (2021)	BERT + Support vector machine and random forest BERT + Gradient boosting regression	85.4%	–	–	–	4.56
Zhu et al. (2021a)	YAMNet + BERT base + Concat/joint feature	82.92%	83.04%	83.33%	82.92%	4.06
Vats et al. (2021)	BERT + KNN + Score fusion	85.2%	–	–	–	–
Valsaraj et al. (2021)	BERT model + TF-IDF	76.06%	80%	73.68%	76.71%	–
Syed et al. (2021)	LR + SVM (majority voting) classification SVM + PLSR (averaging) regression	91.67%	–	91.67%	–	3.74
Meghanani et al. (2021)	CNN-LSTM + pBLSTM-CNN	64.58%	82%	38%	51%	5.9
Martinc et al. (2021)	ADR audio/textual features + top three classifiers late fusion	93.75%	–	–	–	–

Table 5 (continued)

Publication	ML/DL models	Accuracy	Precision	Recall	F1	RMSE
Mahajan and Baths (2021)	Bi-modal model (GRU + CNN + Bi-LSTM + attention Layer)	72.92%	78.94%	62.5%	69.76%	–
Li and Huang (2021)	1dCNN-triplet + DT	83.3%	88%	84%	85%	–
Balagopalan et al. (2021)	BERT, Ridge	83.3%	83.8%	83.3%	83.3%	4.56
Chlasta and Wolk (2021)	VGGish + 128 PCA + DemCNN	62.5%	62.5%	62.5%	62.5%	–
Shah et al. (2021)	Majority vote (NLP + Acoustic); Random forest (NLP)	81%	84%	81%	81%	6.43
Searle et al. (2020)	DistilBERT	81%	80%	83%	82%	4.58
Pappagari et al. (2020)	x-vector + Probabilistic linear discriminant analysis (PLDA) + SVR	75%	70%	88%	78%	5.37
Syed et al. (2020)	BERT + Score fusion					
	label fusion from the top-10 performing models BEERT	85.42%	–	–	–	4.3
Sarawgi et al. (2020)	MLP + ensemble	83%	83%	83%	83%	4.6
Pompili et al. (2020b)	i-vectors, x-vectors + BERT + LSTM-RNNs with attention + Fusion	81.25%	94.12%	66.67%	78.05%	–
Koo et al. (2020)	VGGish, Pre-trained language model features + CRNN	81.25%	89.47%	70.83%	79.07%	3.7
Cummins et al. (2020)	BoAW-MFCC + bi-LSTM with attention + Fusion majority vote	85.2%	–	–	85.4%	4.65
Edwards et al. (2020)	FastText + LDA (ComPareE2016)	79.17%	81.82%	75%	78.26%	–
Martinc and Pollak (2020)	Early fusion + grid search + SVM	77.08%	–	–	–	4.43
Rohanian et al. (2020)	LSTM with gating (acoustic + Lexical + Dis)	79.17%	81.82%	75%	78.26%	4.54
Yuan et al. (2020)	ERNIE + Pause feature	89.6%	95.2%	83.3%	88.9%	–
Balagopalan et al. (2020)	BEERT + Ridge	83.3%	81%	88%	84%	4.56

Luz et al. (2020) described the ADReSS Challenge in detail and presented a baseline for comprehensive methodological comparisons

1. **Comparability:** standardized datasets and evaluation metrics have enabled direct comparisons between different approaches. This comparability has facilitated a more robust assessment of methodological advancements, as evidenced by the progressive improvements in accuracy and other performance metrics seen in Tables 5 and 6.
2. **Reproducibility:** by using common datasets, researchers can more easily reproduce and validate each other's work, enhancing the reliability of findings in the field. This has led to a more solid foundation for further research and development.
3. **Focus on real-world applicability:** the ADReSSo challenge has pushed researchers to develop models that work directly with speech input, closely aligning with real-world

Table 6 Comparisons of research on ADReSSo dataset

Publication	ML/DL Models	Accuracy	Precision	Recall	F1	RMSE
Luz et al. (2021) Baseline	late fusion + SVM, support vector regression (SVR)	78.87%	77.8%	80%	78.87%	5.29
Ilias and Askounis (2023)	BERT + DeiT + Optimal transport + Co-attention fusion (deep context) + label smoothing	85.35%	84.43%	86.29%	85.27%	–
Priyadarshinee et al. (2023)	Otter.ai transcription + RoBERTa + DNN (3-layers)	88.7%	–	–	–	–
Pan et al. (2023)	WAV2VEC2.0 + Tree bagger (TB)	78.87%	–	–	78.49%	–
Liu et al. (2023)	VAD Pause feature and eGeMAPS + Tree bagger (TB)	70.7%	–	–	–	–
Yadav et al. (2022)	Multi-layer Perceptron	71.85%	–	–	–	–
Vats et al. (2022)	Linear support vector machine	63.4%	–	–	–	–
Deng et al. (2022)	SVM + multi-head-self-attention (MHSA)-CNN + Average fusion	87.32%	87.62%	87.26%	87.28%	–
Agbavor and Liang (2022)	Wav2Vec2 + GPT-3 Text embedding + SVM	80.3%	72.3%	97.1%	82.9%	6.25
Rohanian et al. (2021)	LSTM w/Gating	84%	–	–	–	4.26
Pan et al. (2021)	<i>BERT_{large}</i>	84.51%	81.58%	88.57%	84.93%	–
Zhu et al. (2021b)	WavBERT	83.1%	87.1%	77.14%	81.82%	4.44
Pappagari et al. (2021)	logistic regression + Score fusion	84.51%	92%	74%	83%	3.85
Pandey et al. (2021)	CNN + DistillBert + Score fusion	73.68%	69.69%	82.14%	75.4%	–
Liu et al. (2021b)	MLP	97.18%	–	96.34%	97.09%	3.76

Luz et al. (2021) described the ADReSSo Challenge in detail and presented a baseline for comprehensive methodological comparisons

clinical applications. This focus has accelerated the development of techniques that could potentially be translated into practical diagnostic tools.

6 Discussion

In this section, we aggregate findings from various studies to provide a comprehensive view of the landscape of Alzheimer's disease detection through speech analysis. This discussion allows for a nuanced understanding of the methodologies and trends within the field.

6.1 Domain-specific features versus deep learning-derived features

The debate over incorporating domain-specific features versus relying on deep learning-derived features in feature engineering is a critical issue in the field of AD detection through speech analysis.

Traditional models with domain-specific features: Traditional models leverage domain-specific features, which are manually crafted based on expert knowledge in linguistics and acoustics. These features enhance model interpretability, contributing to greater transparency in decision-making processes. For instance, linguistic features might include syntax, semantics, and phonetic patterns, while acoustic features might encompass pitch, tone, and pauses. The primary advantage of this approach is its interpretability, as clinicians can easily understand and trust the model's decision-making process. However, the downside is that the model's performance is heavily reliant on the quality and comprehensiveness of the manually extracted features, which may not capture all the nuances present in the data.

Deep learning-derived features: In contrast, deep learning models, such as those utilizing BERT embeddings, automatically derive features from raw data, often capturing complex patterns and long-range dependencies that domain-specific features might miss. Deep learning models have achieved state-of-the-art performance across various NLP tasks due to their ability to model intricate relationships within the data. For example, Balagopalan et al. (2021) compared two approaches for the automatic detection of AD from speech: one using domain knowledge to extract linguistic and acoustic features, and another using pre-trained language models like BERT for transfer learning. Both approaches significantly outperformed the baseline linguistic model on the ADReSS dataset, with BERT models showing slightly higher performance, although the difference was not statistically significant. The authors also explored the interpretability of the BERT model, finding that attention weights reveal patterns related to important information content units, pauses, and filler words in the speech.

6.2 Acoustic and linguistic features

In studies focusing solely on audio features, such as that conducted by Meghanani et al. (2021), which utilized log-Mel spectrogram and Mel-Frequency Cepstral Coefficients, the achieved accuracy often decreases to below 70%, as evidenced by an accuracy of 64.58%. Conversely, the efficacy of language models surpasses that of acoustic models in dementia detection, a trend observed in various studies, including the ADReSS Challenge paper (Luz et al. 2020). Additionally, the integration of both acoustic and linguistic features has been shown to provide complementary information, as highlighted in the literature (Deng et al. 2022), with most studies (Meerza et al. 2022; Zhu et al. 2021a; Deng et al. 2022) employing this combined approach achieving accuracies above 85%. Notably, Parsapoor et al. (2023) are among the few to suggest that machine learning classifiers trained with acoustic features outperform those trained solely with linguistic features.

6.3 Dominance of deep learning and pre-trained models

A prevailing trend in the literature is the widespread utilization of deep learning models (Priyadarshinee et al. 2023; Pan et al. 2021; Ilias et al. 2023; Chen et al. 2023b; Chatzianastasis

et al. 2023; Zhu et al. 2021b; Vats et al. 2022; Pérez-Toro et al. 2022; Pandey et al. 2021), often leveraging pre-trained models such as BERT. This underscores the effectiveness and popularity of these advanced techniques compared with Domain-specific features in capturing intricate patterns in speech and transcript data, showcasing their dominance in the field.

6.4 Temporal features for AD detection

Notably, features related to pauses and silence have been identified as valuable contributors to AD detection. MohamedShreif and Lawgali (2022), Liu et al. (2023) and Rohanian et al. (2021) findings highlight the complementary information provided by these temporal features, suggesting their significance in capturing nuances associated with cognitive impairments in speech patterns. Focusing solely on accuracy, Martinc et al. (2021) demonstrated the efficacy of employing semantic bag-of-n-grams features alone, achieving an accuracy of 89.58% in discerning between individuals with AD and healthy controls. Subsequent incorporation of temporal and structural information, achieved through the integration of bag-of-n-grams features with ADR audio/textual features, led to an enhancement in accuracy, reaching 91.67% on the test dataset. Notably, employing late fusion techniques yielded an accuracy of 93.75%, representing the highest reported performance in the literature for ADReSS dataset.

6.5 Exploration of non-speech representations

Syed et al. (2021) proposed the utilization of deep acoustic embeddings derived from environmental sounds and music for dementia classification and regression tasks. Their suggestion of the potential utility of non-speech representations in speech-related applications emphasizes the need to explore diverse sources of information beyond traditional speech features, pointing towards a broader perspective in feature extraction for dementia recognition.

7 Challenge

Despite extensive research over many years, recognizing AD through speech remains challenging. The key challenges are highlighted in the following subsections, with relevant facts to provide a clearer understanding of the issues.

7.1 Data scarcity

The main problem of data scarcity is overfitting. Berisha et al. (2022) showed that reported accuracy declines as sample size increases, indicating an overoptimistic estimate of accuracy in small sample size studies. The overestimation of true model accuracy in published literature can lead to unrealistic expectations and potential negative consequences in the deployment of clinical speech analytic models. There are three potential solutions to data scarcity.

The first approach to address this challenge involves proactively addressing dataset imbalances by collecting new data. However, it is important to acknowledge that this under-

taking may entail considerable time, financial investment, and resource allocation, making it a potentially arduous task for individuals unless collaborative efforts with external entities, such as healthcare industry stakeholders, are initiated.

The second approach is to use transfer learning technique by training the model with a large dataset comprising of normal speech. Yang et al. (2022) proposed an augmented adversarial self-supervised learning method for AD detection using limited speech data. The model is trained in an adversarial manner using limited Alzheimer's data with a large scale of easily-collected normal speech data and an augmented set of Alzheimer's data. This could be a more practical, easy and effective approach to handle the data sparsity problems.

The third approach is to use data augmentation technique. Data augmentation is a common method for expanding the amount of training data, avoiding over-fitting, and boosting model robustness. Woszczyk et al. (2022) used Noise, Lexical substitution, Paraphrase, and Text Generation for text augmentation and Standard transformations, Vocal Tract Length Perturbation and generative models for audio augmentation. Their results showed that data augmentation improves the performance of the models, achieving comparable results to state-of-the-art models on the ADReSS dataset. Bertini et al. (2021) pioneered the application of SpecAugment, a data augmentation technique, which operates on the log mel spectrogram of the input audio. SpecAugment consists of three operations: time warping, frequency masking, and time masking. These operations are applied to each input spectrogram, resulting in a final training dataset twice the size of the original one. Moreover, This approach is computationally cheaper compared to methods based on audio deformation.

7.2 Lack of standardized datasets

With recent advancements in AD research, the emergence of challenges such as ADReSS (Luz et al. 2020), ADReSSo (Luz et al. 2021), and ADReSS-m (Luz et al. 2023) has led to the creation of more benchmark datasets. However, it is noteworthy that a significant portion of these challenge datasets are derived from the DementiaBank resource, resulting in limited diversity in data sources. There exists a notable disparity in terms of dataset availability across different modalities, such as neuroimaging, which are essential for comprehensive AD research. Furthermore, a pressing concern in the realm of standardization pertains to poor replicability. Not all researchers share their code publicly, exacerbating the challenge of reproducing research findings and hindering the validation of results. Additionally, there is a significant disparity in the availability of datasets across different languages. Most of the existing datasets are predominantly in English, limiting the generalizability of research findings to non-English-speaking populations. This linguistic limitation can result in biased models that may not perform effectively across diverse linguistic backgrounds, underscoring the need for multilingual datasets to enhance the robustness and applicability of AD detection methods globally.

7.3 Privacy concerns

The integration of machine learning in healthcare raises significant ethical and privacy concerns that demand careful consideration and mitigation strategies. These concerns encompass various aspects such as data usage and patient confidentiality. One approach to address

privacy concern is to adopt federated learning which was first proposed by Google in 2016 (McMahan et al. 2017).

Meerza et al. (2022) proposed the first federated learning-based approach for automatic AD diagnosis through speech analysis, ensuring privacy and fairness. Their machine learning model consisted of a Long Short-term Memory network and a feed-forward neural network based on combination of acoustic mel-frequency features and linguistic features (pause rate and duration). This model achieved high accuracy 89.1% in centralized learning and comparable performance in federated learning settings. However, there is no performance guarantee for each client due to the data heterogeneity. For example, some patients with sufficient data may have a relatively high accuracy than others.

7.4 Interpretable concerns

Interpretable is also a major concern in the healthcare system. It allows for reliable explanations of automated AD diagnosis, which is crucial for clinical impact and decision-making. First, interpretability helps build trust in machine learning models among healthcare professionals, patients, and regulatory bodies. When stakeholders can understand and interpret the decisions made by a model, they are more likely to adopt and accept it as a valuable tool in healthcare decision-making. Second, interpretability is essential for clinicians to comprehend and trust the recommendations provided by these models, allowing them to make informed decisions about patient diagnosis, treatment plans, and overall management. However, it is against the nature of most deep learning methods used in this field. Because of the “Black Box”, decision-making transparency is very hard to get from normal deep learning methods. The work of Ilias and Askounis (2022a) stands out for its focus on interpretable methods in Alzheimer’s disease detection. The authors achieved interpretability through multiple approaches, including the use of transformer-based models, siamese networks, and explainable AI techniques such as Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al. 2016). They emphasized model interpretability to reveal the linguistic patterns and differences between AD and non-AD patients. Their multi-task learning model, which simultaneously detects AD and assesses its severity, further contributes to the interpretability by providing a more comprehensive analysis. The authors’ detailed linguistic analysis, including text statistics, vocabulary uniqueness, and word usage correlations, offers additional interpretable insights into the language characteristics of AD patients in deep learning.

8 Future direction

8.1 Benchmarking with previous models

Current approaches to benchmarking AD detection models often face challenges since they utilize different datasets, making direct comparisons more complex. A critical future direction is the re-implementation of previous best-performing methods on standardized datasets such as the ADReSS benchmark for meaningful comparison. Mahajan and Baths (2021) attempted this by re-implementing existing NLP methods using CNN-LSTM architectures on the ADReSS dataset, finding that accuracy dropped to 72.92%, in contrast to state-of-the-

art results on the DementiaBank dataset. The recurring instances of samples from the same subject within the Pitt Corpus dataset constitute a predominant factor. Such recurrent occurrences have the potential to induce substantial overfitting in models developed utilizing this dataset, particularly with respect to participant-specific characteristics.

8.2 Practical application of AD models

Although state-of-the-art AD models have shown promise, their practical application in clinical settings requires further rigorous testing and implementation. In this context, Pigliautile et al. (2022) pioneered the development of the Dementia Monitoring application (DMapp) as an integral component of the Memento project. This innovative application harnesses recent advancements in Natural Language Processing techniques for meticulous textual analysis. Moreover, extending the scope beyond clinical settings, Liang et al. (2022) focused on early cognitive decline detection among older adults using voice commands from Amazon Alexa. Machine-learning models, including decision trees and random forests, achieved high classification accuracies ranging from 80 to 90%. However, these applications highlight the need for further development to ensure reliability and accuracy in diverse real-world scenarios. Future research should focus on refining these models for broader and more consistent clinical and home-based use.

8.3 Multilingual AD recognition

The use of multilingual models offers a practical solution to the problem of lacking of a large dataset in many languages. As there is a limited collection of text data from Alzheimer's patients in many languages, training a multilingual model in a source language (such as English) and applying it to making inference in the target language can offer a valuable solution. The ADRess-M Challenge 2023 (Luz et al. 2023) exemplifies this by having

Table 7 Comparisons of research on ADRess-M dataset

Publication	ML/DL models	Accuracy	Precision	Recall	F1	RMSE
Shah et al. (2023)	Logistic regression with L2-regularizer Support vector machine	69.57%	—	—	—	4.7693
Mei et al. (2023)	XGBoost employed for AD detection Support vector regression (SVR) and XGBoost regressors	73.9%	—	—	—	4.610
Chen et al. (2023b)	OpenSmile tool + SVM XLSR-53 + fully connected layer (FC) Whisper + RoBERTa + FC	69.6%	69.23%	75%	72%	4.788
Tamm et al. (2023)	25-D eGeMAPS feature vector (openSMILE) batch normalization + down-projected feature + attention pooling + linear projection	82.6%	88.9%	72.7%	80%	4.345
Jin et al. (2023)	wav2vec, x-vector feature, 18 disfluency features Swin transformer, random forests, DNN, gradient boosting Complementary and simultaneous ensemble, selective averaging, majority voting	86.96%	—	—	—	3.727

participants train models on English speech data and evaluate them on Greek speech data, aiming to identify speech features that are predictive across languages. Table 7 presents a comparison of research conducted on the ADReSS-M dataset. However, significant research gaps still remain for languages with substantial linguistic differences from English. Future research should aim to develop universal models that can effectively generalize across more diverse languages and dialects, incorporating culturally and linguistically relevant features. Here are several studies that specifically examine non-English languages.

Pérez-Toro et al. (2022) investigated the feasibility of combining information from English and Spanish languages to discriminate AD which based on known acoustic (Wav2Vec) and word (BERT, RoBERTa) embeddings using different classifiers.

Bertini et al. (2022b) proposed a cross-language AD classifier based on spontaneous speech, which is an end-to-end autoencoder model trained on spectrogram. Their method achieved good classification results for both English and Italian languages which indicates its ability to capture language-independent signs of dementia.

Lindsay et al. (2021) includes participants from two different languages (English and French) and extracts task-specific, semantic, syntactic, and paralinguistic features using NLP technique to detect language impairment in AD through picture description tasks. They find out the generalizable language feature set, particularly semantic features, outperforms the all language feature set in detecting language impairment in AD, showing its effectiveness in clinical applications.

Adhikari et al. (2022) created a novel dataset of transcripts from AD patients and control subjects in Nepali and applied various ML and DL algorithms for classification. The author suggested further improvements, namely including acoustic features such as the duration of pause a person takes while speaking.

Sangchocanonta et al. (2021) presents the development of a Thai Picture Description Task for Alzheimer's screening using PoS tagging. Two picture description tasks, Thais-at-Home and Thai Temple Fair, were designed to be suitable and relatable to Thai culture. Six machine learning algorithms (Adaboost, Logistic regression, Multilayer perceptron, Native Bayes, Random Forest, and Decision tree) were applied to train with the PoS feature, and the multilayer perceptron model achieved the best classification results with 90.00% sensitivity, 80.00% specificity, and 86.67% accuracy in distinguishing patients with AD and mild cognitive impairment from healthy controls.

Calzà et al. (2021) present a pioneering investigation, as it constitutes the first known study conducted in the Italian language context. It examined an extensive array of features for the purpose of constructing automated classifiers designed to identify individuals with mild cognitive impairment from health control. A multidimensional parameters computation was performed on the data, taking into consideration a large set of 87 acoustical, rhythmical, morpho-syntactic, and lexical features as well as some readability indexes and demographic information. Subsequently, the Authors used two different machine learning classifiers, Support Vector Classifier and the Random Forest Classifier. Notably, SVC achieves high F1 scores around 75%, underscoring its efficacy as a valuable tool for the classification task. However, the author chose not to employ any DNN classifier as they did not believe there was enough annotated data to train them.

Bertini et al. (2021) trained an autoencoder using the spectrogram representation of the audio signal and used a data augmentation approach to overcome the lack of annotated data in the Italian language. The learned representations of each spectrogram are extracted from

the hidden layer of the autoencoder and fed into a multilayer perceptron for classification, such proposed method achieves an accuracy of 90.57% in classifying the audio files of subjects with MCI and early dementia.

8.4 Emerging multimodal approaches in dementia detection

Recent research has begun exploring the integration of multiple modalities beyond speech and text, such as eye-tracking, gait, and drawing behaviors, to enhance dementia detection. Sheng et al. (2022) propose a method of detecting dementia from the simultaneous speech and eye-tracking recordings of subjects in a picture description task. Their experimental results demonstrate that the detection accuracy of the proposed method is 84.26%, which outperforms baseline methods and ablated models using single speech or eye-tracking input.

Similarly, Jang et al. (2021) analyses four tasks: pupil fixation, description of a pleasant past experience, picture description, and paragraph reading. The novel tasks (pupil fixation and pleasant past experience) showed similar classification accuracy to the established tasks (picture description and paragraph reading), indicating their discriminative ability for memory clinic patients. The fusion of multimodal data across tasks yielded the highest overall AUC of 0.83 ± 0.01 .

Apart from eye-tracking, Yamada et al. (2021) investigated the use of gait, speech, and drawing behaviors as potential indicators for the diagnosis of AD and mild cognitive impairment. Combining all three behavioral modalities achieved 93.0% accuracy in classifying AD, MCI, and CN, compared to 81.9% accuracy when using the best individual modality. Each behavioral modality was associated with different cognitive and clinical measures for diagnosing AD and MCI, indicating that they provide complementary information about cognitive impairments.

These studies demonstrate the current application and potential of multimodal approaches in providing complementary information for dementia detection. While showing promise, there's room for further exploration. Future research could focus on integrating additional modalities and refining fusion techniques to enhance diagnostic accuracy and robustness across diverse populations.

8.5 Automatic speech recognition (ASR) in AD

Current commercial ASR systems often exhibit high word error rates (WER) when used with AD patients, impacting the accuracy of downstream tasks. Despite various efforts using systems like Google Cloud Speech Recognizer, IBM Watson, and iFlyt, transcription errors remain a significant challenge. Future research should focus on developing ASR models tailored specifically for AD patients' unique speech patterns. This could involve using confidence scores as a measure of transcription quality and integrating these scores into subsequent analyses to mitigate the impact of errors or incorporating ASR systems directly into the end-to-end training loop. Here are some studies that utilize different existing ASR system for downstream AD detection task.

In the ADReSSo challenge, Luz et al. (2021) utilized the Google Cloud-based Speech Recognizer to automatically transcribe audio files, converting these transcripts into the CHAT format for linguistic analysis. Calzà et al. (2021) also employed Google Cloud ASR, trained for Italian, achieving a WER of 27.78%, which was considered acceptable given the

nature of the speech samples. Pompili et al. (2020a) used the Google Cloud Speech-to-Text API, acknowledging that erroneous transcriptions could impact AD detection performance. To address this, Pan et al. (2021) incorporated ASR confidence scores into text embeddings to inform the classifier about transcription quality. Soroski et al. (2022) compared Google Cloud STT-generated transcripts with manually verified ones, examining confidence scores, error rates, and classification accuracy for AD.

Apart from Google ASR system, Yamada et al. (2023) used IBM Watson for automatic transcription in their research. Similarly, Rohanian et al. (2021) also employed IBM Watson, leveraging its detailed word timing outputs as features in their system. This highlighting the importance of IBM Watson ASR due to its retention of hesitation markers and disfluencies, which are crucial features for AD detection. Deng et al. (2022) used iFlyt ASR to obtain transcripts for the ADReSSo challenge, noting that despite some accuracy loss due to transcription errors, their system remained robust for AD recognition. Liu et al. (2021b) compared two transcript-driven speech recognition systems, CMUSphinx and Mozilla Deep Speech, demonstrating outstanding performance. They hypothesized that ASR-transcribed speech emphasizes hidden features in words and language fluency, potentially affecting AD diagnosis.

Beside previous commercial ASR system, Ye et al. (2021) developed a state-of-the-art ASR system using the DementiaBank Pitt corpus, incorporating techniques like segmentation refinement, audio augmentation, Bayesian adaptation, and a Transformer language model. This system reduced the WER by 11.72% absolute (26.11% relative) over the baseline system, with future research directions including tighter integration of recognition and NCD detection components, fusion with paralinguistic features, and further perturbation of elderly speech.

9 Conclusion

The utilization of speech data in the detection of AD represents a promising avenue for clinical diagnosis and decision-making. As AD continues to present a growing healthcare challenge worldwide, the importance of non-invasive, accessible diagnostic tools cannot be overstated. This review paper highlights the advancement of artificial intelligence and machine learning algorithms in detecting and understanding the relationship between speech patterns and AD.

The major developments in the field are emphasized via a comprehensive analysis of state-of-the-art algorithms and a rigorous comparative examination using benchmark datasets such as ADReSS and ADReSSo. These advancements not only underscore the potential of speech-based diagnostic approaches but also offer valuable insights for future research and clinical applications. Moreover, we also elaborate on several challenges that still persist in this domain, including benchmark issues related to data scarcity, poor standardization, privacy, and interpretive concerns.

The findings from this review have significant potential clinical implications. The high accuracy rates achieved by integrating acoustic and linguistic features, particularly through deep learning models, suggest that speech-based AD detection could become a valuable tool in clinical practice. This non-invasive, cost-effective method could enable earlier detection and intervention, potentially improving patient outcomes and quality of life. However,

translating these findings into practice requires addressing several challenges, including the development of user-friendly, interpretable tools for clinicians, and the refinement of multilingual models to expand access to diverse populations, while privacy-preserving techniques like federated learning could facilitate larger-scale implementation across healthcare systems. Additionally, the exploration of multimodal approaches combining speech with other behavioural markers (e.g., eye-tracking, gait analysis) could further enhance diagnostic accuracy.

Despite these promising developments, this review acknowledges several limitations inherent in the current body of research. Publication bias may skew the perceived efficacy of certain methods, as studies with negative or null results are less likely to be published. The search strategy and selection criteria for included studies could have inadvertently omitted relevant research, leading to an incomplete synthesis of the literature. Also, Limiting the review to studies published within five year can exclude older studies that may still be relevant. The quality of the included studies varies, which may impact the reliability of the conclusions drawn. Additionally, the rapidly evolving nature of this field means that our review might quickly become outdated as new studies are published. This survey will enable future research to refine and expand upon existing methods, enhance opportunities for collaboration across interdisciplinary domains, and further investigate the potential of multi-modal approaches that combine speech data with other clinical indicators. This holistic view of the domain will culminate in the development of more accurate and accessible diagnostic tools for AD.

As the review provides a unique comparison of different AI methods with the same benchmark datasets, it will serve as a catalyst for translating crucial research findings into meaningful clinical outcomes. As we continue to explore the intricate relationship between speech and AD, these efforts of systematically consolidating information will ultimately enhance early detection, intervention, and care for individuals affected by this debilitating condition. While promising, it's important to note that these speech-based methods should complement, not replace, existing diagnostic procedures, with their primary value lying in early screening and monitoring. As research progresses, clinical trials will be necessary to validate these approaches and establish standardized protocols for their use in practice.

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Declarations

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References

- Ablimit A, Botelho C, Abad A et al (2022) Exploring dementia detection from speech: cross corpus analysis. In: ICASSP 2022—2022 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 6472–6476
- Adhikari S, Thapa S, Naseem U et al (2022) Exploiting linguistic information from Nepali transcripts for early detection of Alzheimer's disease using natural language processing and machine learning techniques. *Int J Hum Comput Stud* 160:102761
- Agbavor F, Liang H (2022) Predicting dementia from spontaneous speech using large language models. *PLOS Digital Health* 1(12):e0000168
- Alkenani AH, Li Y, Xu Y et al (2021) Predicting Alzheimer's disease from spoken and written language using fusion-based stacked generalization. *J Biomed Inform* 118:103803
- Association Alzheimer's (2023) 2023 Alzheimer's disease facts and figures. *Alzheimer's Dementia* 19(4):1598–1695. <https://doi.org/10.1002/alz.13016>
- Aryal SK, Prioleau H, Burge L (2022) Acoustic-linguistic features for modeling neurological task score in Alzheimer's. In: Pacific symposium on biocomputing 2023: Kohala Coast, Hawaii, USA, 3–7 January 2023. World Scientific, pp 335–346
- Baevski A, Zhou Y, Mohamed A et al (2020) wav2vec 2.0: A framework for self-supervised learning of speech representations. *Adv Neural Inf Process Syst* 33:12449–12460
- Balogopalan A, Eyre B, Rudzicz F et al (2020) To BERT or not to BERT: comparing speech and language-based approaches for Alzheimer's disease detection. *arXiv preprint arXiv:2008.01551*
- Balogopalan A, Eyre B, Robin J et al (2021) Comparing pre-trained and feature-based models for prediction of Alzheimer's disease based on speech. *Front Aging Neurosci* 13:635945
- Bank D, Koenigstein N, Giryres R (2023) Autoencoders, machine learning for data science handbook: data mining and knowledge discovery handbook, pp 353–374
- Becker JT, Boiler F, Lopez OL et al (1994) The natural history of Alzheimer's disease: description of study cohort and accuracy of diagnosis. *Arch Neurol* 51(6):585–594
- Begam B, Palanivelan M, Preethi S (2023) An ensemble machine learning algorithm to diagnose Alzheimer's disease. In: 2023 International conference on recent advances in electrical, electronics, ubiquitous communication, and computational intelligence (RAEEUCCI). IEEE, pp 1–6
- Berisha V, Krantsevich C, Stegmann G et al (2022) Are reported accuracies in the clinical speech machine learning literature overoptimistic? In: Proceedings of the annual conference of the international speech communication association. INTERSPEECH, pp 2453–2457
- Bertini F, Allevi D, Lutero G et al (2021) Automatic speech classifier for mild cognitive impairment and early dementia. *ACM Trans Comput Healthc (HEALTH)* 3(1):1–11
- Bertini F, Allevi D, Lutero G et al (2022a) An automatic Alzheimer's disease classifier based on spontaneous spoken English. *Comput Speech Lang* 72:101298
- Bertini F, Allevi D, Lutero G et al (2022) A cross-language dementia classifier: a preliminary study. In: 2022 IEEE international conference on metrology for extended reality, artificial intelligence and neural engineering (MetroXRaine). IEEE, pp 438–443
- Boletsis C (2020) A review of automated speech-based interaction for cognitive screening. *Multimodal Technol Interact* 4(4):93
- Bouazizi M, Zheng C, Ohtsuki T (2022) Dementia detection using language models and transfer learning. In: 2022 The 5th international conference on software engineering and information management (ICSIM), pp 152–157
- Breijyeh Z, Karaman R (2020) Comprehensive review on Alzheimer's disease: causes and treatment. *Molecules* 25(24):5789
- Burke E, Gunstad J, Pavlenko O et al (2023) Distinguishable features of spontaneous speech in Alzheimer's clinical syndrome and healthy controls. *Aging Neuropsychol Cogn* 1–12
- Calzà L, Gagliardi G, Favretti RR et al (2021) Linguistic features and automatic classifiers for identifying mild cognitive impairment and dementia. *Comput Speech Lang* 65:101113
- Chatzianastasis M, Ilias L, Askounis D et al (2023) Neural architecture search with multimodal fusion methods for diagnosing dementia. In: ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 1–5

- Chen W, Xing X, Xu X et al (2023a) Speechformer++: a hierarchical efficient framework for paralinguistic speech processing. *IEEE/ACM Trans Audio Speech Lang Process* 31:775–788
- Chen X, Pu Y, Li J et al (2023) Cross-lingual Alzheimer's disease detection based on paralinguistic and pre-trained features. In: *ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, pp 1–2
- Chlasta K, Wolk K (2021) Towards computer-based automated screening of dementia through spontaneous speech. *Front Psychol* 11:623237
- Cortes C, Vapnik V (1995) Support-vector networks. *Mach Learn* 20:273–297
- Cummins N, Pan Y, Ren Z et al (2020) A comparison of acoustic and linguistics methodologies for Alzheimer's dementia recognition. In: *Interspeech 2020, ISCA—International speech communication association*, pp 2182–2186
- Deng H, Liu H, Zhou Y et al (2022) Alzheimer's disease detection using acoustic and linguistic features. In: *2022 IEEE 24th Int Conf on High performance computing & communications; 8th Int Conf on data science & systems; 20th Int Conf on Smart City; 8th Int Conf on dependability in sensor, cloud & big data systems & application (HPCC/DSS/SmartCity/DependSys)*. IEEE, pp 2280–2284
- Devlin J, Chang MW, Lee K et al (2018) Bert: pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*
- Dong Z, Zhang Z, Xu W et al (2024) Hafformer: a hierarchical attention-free framework for Alzheimer's disease detection from spontaneous speech. In: *ICASSP 2024—2024 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, pp 11246–11250
- Edwards E, Dognin C, Bollepalli B et al (2020) Multiscale system for Alzheimer's dementia recognition through spontaneous speech. In: *Interspeech*, pp 2197–2201
- Eyben F, Weninger F, Gross F, et al (2013) Recent developments in opensmile, the Munich open-source multimedia feature extractor. In: *Proceedings of the 21st ACM international conference on multimedia*, pp 835–838
- de la Fuente Garcia S, Ritchie CW, Luz S (2020) Artificial intelligence, speech, and language processing approaches to monitoring Alzheimer's disease: a systematic review. *J Alzheimer's Dis JAD* 78(4):1547–1574. <https://doi.org/10.3233/JAD-200888>
- Garcia SDLF, Haider F, Fromm D et al (2024) Connected speech-based cognitive assessment in Chinese and English. *arXiv preprint arXiv:2406.10272*
- Gauthier S, Webster C, Servaes S et al (2022) World Alzheimer report 2022 life after diagnosis: navigating treatment, care and support. <https://www.alzint.org/u/World-Alzheimer-Report-2022.pdf>
- Gkoumas D, Wang B, Tsakalidis A et al (2024) A longitudinal multi-modal dataset for dementia monitoring and diagnosis. *Lang Resour Eval* 1–20
- Goodglass H, Kaplan E, Weintraub S (2001) BDAE: The Boston diagnostic aphasia examination. Lippincott Williams & Wilkins, Philadelphia
- Hajjar I, Okafor M, Choi JD et al (2023) Development of digital voice biomarkers and associations with cognition, cerebrospinal biomarkers, and neural representation in early Alzheimer's disease. *Alzheimer's Dementia Diagn Assess Dis Monit* 15(1):e12393
- Hason L, Krishnan S (2022) Spontaneous speech feature analysis for Alzheimer's disease screening using a random forest classifier. *Front Digital Health* 4:901419
- Haulcy R, Glass J (2021) Classifying Alzheimer's disease using audio and text-based representations of speech. *Front Psychol* 11:624137
- Horigome T, Hino K, Toyoshiba H et al (2022) Identifying neurocognitive disorder using vector representation of free conversation. *Sci Rep* 12(1):12461
- Hsu WN, Bolte B, Tsai YHH et al (2021) Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Trans Audio Speech Lang Process* 29:3451–3460
- Igarashi T, Umeda-Kameyama Y, Kojima T et al (2023) Questionnaires for the assessment of cognitive function secondary to intake interviews in in-hospital work and development and evaluation of a classification model using acoustic features. *Sensors* 23(11):5346
- Ilias L, Askounis D (2022a) Explainable identification of dementia from transcripts using transformer networks. *IEEE J Biomed Health Inform* 26(8):4153–4164
- Ilias L, Askounis D (2022b) Multimodal deep learning models for detecting dementia from speech and transcripts. *Front Aging Neurosci* 14:830943
- Ilias L, Askounis D (2023) Context-aware attention layers coupled with optimal transport domain adaptation and multimodal fusion methods for recognizing dementia from spontaneous speech. *Knowl Based Syst* 277:110834
- Ilias L, Askounis D, Psarras J (2022) A multimodal approach for dementia detection from spontaneous speech with tensor fusion layer. In: *2022 IEEE-EMBS international conference on biomedical and health informatics (BHI)*. IEEE, pp 1–5

- Ilias L, Askounis D, Psarras J (2023) Detecting dementia from speech and transcripts using transformers. *Comput Speech Lang* 79:101485
- Jang H, Soroski T, Rizzo M et al (2021) Classification of Alzheimer's disease leveraging multi-task machine learning analysis of speech and eye-movement data. *Front Hum Neurosci* 15:716670
- Javed A, Dallora AL, Berglund JS et al (2023) Machine learning for dementia prediction: a systematic review and future research directions. *J Med Syst* 47(1):17
- Jin L, Oh Y, Kim H et al (2023) Consen: complementary and simultaneous ensemble for Alzheimer's disease detection and MMSE score prediction. In: ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 1–2
- Karakostas A, Briassouli A, Avgerinakis K et al (2016) The dem@ care experiments and datasets: a technical report. arXiv preprint [arXiv:1701.01142](https://arxiv.org/abs/1701.01142)
- Koo J, Lee JH, Pyo J et al (2020) Exploiting multi-modal features from pre-trained networks for Alzheimer's dementia recognition. arXiv preprint [arXiv:2009.04070](https://arxiv.org/abs/2009.04070)
- Kumar MR, Vekkot S, Lalitha S et al (2022) Dementia detection from speech using machine learning and deep learning architectures. *Sensors* 22(23):9311
- Kurtz E, Zhu Y, Driesse T et al (2023) Early detection of cognitive decline using voice assistant commands. In: ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 1–5
- Lanzi AM, Saylor AK, Fromm D et al (2023) Dementiabank: theoretical rationale, protocol, and illustrative analyses. *Am J Speech Lang Pathol* 32(2):426–438
- Li J, Huang P (2021) Task-oriented feature representation for spontaneous speech of ad patients. In: Bioinformatics research and applications: 17th international symposium, ISBRA 2021, Shenzhen, China, November 26–28, 2021, proceedings 17. Springer, Berlin, pp 46–57
- Li C, Knopman D, Xu W et al (2022a) Gpt-d: inducing dementia-related linguistic anomalies by deliberate degradation of artificial neural language models. In: 60th annual meeting of the association for computational linguistics, ACL 2022. Association for Computational Linguistics (ACL), pp 1866–1877
- Li R, Wang X, Lawler K et al (2022) Applications of artificial intelligence to aid early detection of dementia: a scoping review on current capabilities and future directions. *J Biomed Inform* 127:104030. <https://doi.org/10.1016/j.jbi.2022.104030>, <https://www.sciencedirect.com/science/article/pii/S1532046422000466>
- Liang X, Batsis JA, Zhu Y et al (2022) Evaluating voice-assistant commands for dementia detection. *Comput Speech Lang* 72:101297
- Lin SY, Chang HL, Hwang JJ et al (2022) Automatic audio-based screening system for Alzheimer's disease detection. In: 2022 IEEE international conference on systems, man, and cybernetics (SMC). IEEE, pp 2770–2775
- Lindsay H, Tröger J, König A (2021) Language impairment in Alzheimer's disease-robust and explainable evidence for ad-related deterioration of spontaneous speech through multilingual machine learning. *Front Aging Neurosci* 13:642033
- Liu Z, Lin W, Shi Y et al (2021a) A robustly optimized BERT pre-training approach with post-training. In: China national conference on Chinese computational linguistics. Springer, Berlin, pp 471–484
- Liu Z, Proctor L, Collier PN et al (2021b) Automatic diagnosis and prediction of cognitive decline associated with Alzheimer's dementia through spontaneous speech. In: 2021 IEEE international conference on signal and image processing applications (ICSIPA). IEEE, pp 39–43
- Liu N, Luo K, Yuan Z et al (2022a) A transfer learning method for detecting Alzheimer's disease based on speech and natural language processing. *Front Public Health* 10:772592
- Liu N, Yuan Z, Tang Q (2022b) Improving Alzheimer's disease detection for speech based on feature purification network. *Front Public Health* 9:835960
- Liu J, Fu F, Li L et al (2023) Efficient pause extraction and encode strategy for Alzheimer's disease detection using only acoustic features from spontaneous speech. *Brain Sci* 13(3):477
- Luz S, Haider F, de la Fuente S et al (2020) Alzheimer's dementia recognition through spontaneous speech: the ADReSS challenge. In: Proc. Interspeech 2020. ISCA, pp 2172–2176. <https://doi.org/10.21437/Interspeech.2020-2571>
- Luz S, Haider F, de la Fuente S et al (2021) Detecting cognitive decline using speech only: the addresso challenge. arXiv preprint [arXiv:2104.09356](https://arxiv.org/abs/2104.09356)<https://doi.org/10.1101/2021.03.24.21254263>
- Luz S, Haider F, Fromm D et al (2023) Multilingual Alzheimer's dementia recognition through spontaneous speech: a signal processing grand challenge. arXiv preprint [arXiv:2301.05562](https://arxiv.org/abs/2301.05562)
- Mahajan P, Baths V (2021) Acoustic and language based deep learning approaches for Alzheimer's dementia detection from spontaneous speech. *Front Aging Neurosci* 13:623607
- Martinc M, Pollak S (2020) Tackling the ADReSS challenge: a multimodal approach to the automated recognition of Alzheimer's dementia. In: Interspeech, pp 2157–2161
- Martinc M, Haider F, Pollak S et al (2021) Temporal integration of text transcripts and acoustic features for Alzheimer's diagnosis based on spontaneous speech. *Front Aging Neurosci* 13:642647

- Masrani V, Murray G, Field T et al (2017) Detecting dementia through retrospective analysis of routine blog posts by bloggers with dementia. In: *BioNLP 2017. Association for Computational Linguistics*, Vancouver, Canada, pp 232–237
- Matošević L, Jović A (2022) Accurate detection of dementia from speech transcripts using Roberta model. In: *2022 45th Jubilee international convention on information, communication and electronic technology (MIPRO). IEEE*, pp 1478–1484
- McMahan B, Moore E, Ramage D et al (2017) Communication-efficient learning of deep networks from decentralized data. In: *Artificial intelligence and statistics. PMLR*, pp 1273–1282
- Meerza SIA, Li Z, Liu L et al (2022) Fair and privacy-preserving Alzheimer's disease diagnosis based on spontaneous speech analysis via federated learning. In: *2022 44th Annual international conference of the IEEE engineering in medicine & biology society (EMBC). IEEE*, pp 1362–1365
- Meghanani A, Anoop C, Ramakrishnan A (2021) An exploration of log-mel spectrogram and MFCC features for Alzheimer's dementia recognition from spontaneous speech. In: *2021 IEEE spoken language technology workshop (SLT). IEEE*, pp 670–677
- Mei K, Ding X, Liu Y et al (2023) The USTC system for ADReSS-m challenge. In: *ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE*, pp 1–2
- MohamedShreif HO, Lawgali A (2022) Speech recognition for early detecting Alzheimer's disease by using machine learning algorithms. In: *2022 International conference on engineering & MIS (ICEMIS). IEEE*, pp 1–6
- Nambiar AS, Likhita K, Pujya KS et al (2022) Comparative study of deep classifiers for early dementia detection using speech transcripts. In: *2022 IEEE 19th India council international conference (INDICON). IEEE*, pp 1–6
- Nishikawa K, Hirakawa R, Kawano H et al (2022) System of predicting dementia using transformer based ensemble learning. In: *2022 IEEE international conference on consumer electronics (ICCE). IEEE*, pp 1–3
- Organization WH (2021) Global status report on the public health response to dementia. World Health Organization, UN
- Page MJ, McKenzie JE, Bossuyt PM et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Int J Surg* 88:105906
- Pan Y, Mirheidari B, Harris JM et al (2021) Using the outputs of different automatic speech recognition paradigms for acoustic-and BERT-based Alzheimer's dementia detection through spontaneous speech. In: *Interspeech*, pp 3810–3814
- Pan Y, Lu M, Shi Y et al (2023) A path signature approach for speech-based dementia detection. *IEEE Signal Process Lett*
- Pandey SK, Shekhawat HS, Bhasin S, et al (2021) Alzheimer's dementia recognition using multimodal fusion of speech and text embeddings. In: *International conference on intelligent human computer interaction. Springer, Berlin*, pp 718–728
- Pappagari R, Cho J, Moro-Velazquez L et al (2020) Using state of the art speaker recognition and natural language processing technologies to detect Alzheimer's disease and assess its severity. In: *Interspeech*, pp 2177–2181
- Pappagari R, Cho J, Joshi S et al (2021) Automatic detection and assessment of Alzheimer disease using speech and language technologies in low-resource scenarios. In: *Interspeech*, pp 3825–3829
- Parsapoor M, Alam MR, Mihailidis A (2023) Performance of machine learning algorithms for dementia assessment: impacts of language tasks, recording media, and modalities. *BMC Med Inform Decis Mak* 23(1):45
- Pérez-Toro PA, Klumpp P, Hernandez A et al (2022) Alzheimer's detection from English to Spanish using acoustic and linguistic embeddings. In: *Proceedings of the Interspeech*, pp 2483–2487
- Pigliautile M, Colombo M, Pizzuti T et al (2022) Dmapp: a developing promising approach to monitor symptoms progression and stimulate memory in Italian people with cognitive impairments. *Aging Clin Exp Res* 34(11):2721–2731
- Pompili A, Abad A, de Matos DM et al (2020a) Pragmatic aspects of discourse production for the automatic identification of Alzheimer's disease. *IEEE J Sel Top in Signal Process* 14(2):261–271
- Pompili A, Rolland T, Abad A (2020b) The inesc-id multi-modal system for the ADReSS 2020 challenge. *arXiv preprint arXiv:2005.14646*
- Pope C, Davis BH (2011) Finding a balance: the carolinas conversation collection. *Corpus Linguist Linguist Theory* 7(1):143–161
- Priyadarshinee P, Clarke CJ, Melechovsky J et al (2023) Alzheimer's dementia speech (audio vs. text): multi-modal machine learning at high vs. low resolution. *Appl Sci* 13(7):4244
- Qian C, Di J, Li J (2022) Detecting mild cognitive impairment in Alzheimer's disease using speech acoustics only: a two-stage deep metric learning approach. In: *2022 IEEE international conference on bioinformatics and biomedicine (BIBM). IEEE*, pp 2682–2689

- Ribeiro MT, Singh S, Guestrin C (2016) “Why should I trust you?” explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, pp 1135–1144
- Rohanian M, Hough J, Purver M (2020) Multi-modal fusion with gating using audio, lexical and disfluency features for Alzheimer’s dementia recognition from spontaneous speech. ISCA
- Rohanian M, Hough J, Purver M (2021) Alzheimer’s dementia recognition using acoustic, lexical, disfluency and speech pause features robust to noisy inputs. arXiv preprint [arXiv:2106.15684](https://arxiv.org/abs/2106.15684)
- Roshanzamir A, Aghajan H, Soleymani Baghshah M (2021) Transformer-based deep neural network language models for Alzheimer’s disease risk assessment from targeted speech. BMC Med Inform Decis Mak 21:1–14
- Sangchocanonta S, Vongsurakrai S, Sroykhumpa K et al (2021) Development of Thai picture description task for Alzheimer’s screening using part-of-speech tagging. In: 2021 43rd Annual international conference of the IEEE engineering in medicine & biology society (EMBC). IEEE, pp 2104–2109
- Sanh V, Debut L, Chaumond J et al (2019) Distilbert, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint [arXiv:1910.01108](https://arxiv.org/abs/1910.01108)
- Sarawgi U, Zulfikar W, Soliman N et al (2020) Multimodal inductive transfer learning for detection of Alzheimer’s dementia and its severity. arXiv preprint [arXiv:2009.00700](https://arxiv.org/abs/2009.00700)
- Scheltens P, De Strooper B, Kivipelto M et al (2021) Alzheimer’s disease. Lancet 397(10284):1577–1590
- Searle T, Ibrahim Z, Dobson R (2020) Comparing natural language processing techniques for Alzheimer’s dementia prediction in spontaneous speech. arXiv preprint [arXiv:2006.07358](https://arxiv.org/abs/2006.07358)
- Shah Z, Sawalha J, Tasnim M et al (2021) Learning language and acoustic models for identifying Alzheimer’s dementia from speech. Front Comput Sci 3:624659
- Shah Z, Qi SA, Wang F et al (2023) Exploring language-agnostic speech representations using domain knowledge for detecting Alzheimer’s dementia. ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 1–2
- Sheng Z, Guo Z, Li X et al (2022) Dementia detection by fusing speech and eye-tracking representation. In: ICASSP 2022—2022 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 6457–6461
- Soroski T, da Cunha Vasco T, Newton-Mason S et al (2022) Evaluating web-based automatic transcription for Alzheimer speech data: transcript comparison and machine learning analysis. JMIR Aging 5(3):e33460
- Syed MSS, Syed ZS, Lech M et al (2020) Automated screening for Alzheimer’s dementia through spontaneous speech. In: Interspeech, pp 2222–6
- Syed ZS, Syed MSS, Lech M et al (2021) Automated recognition of Alzheimer’s dementia using bag-of-deep-features and model ensembling. IEEE Access 9:88377–88390
- Tamm B, Vandenberghe R, Van Hamme H (2023) Cross-lingual transfer learning for Alzheimer’s detection from spontaneous speech. In: ICASSP 2023—2023 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 1–2
- Thaler F, Gewald H (2021) Language characteristics supporting early Alzheimer’s diagnosis through machine learning—a literature review. Health Inform Int J 10(1):5–23
- Tripathi A, Chakraborty R, Koppurapu SK (2021) Dementia classification using acoustic descriptors derived from subsampled signals. In: 2020 28th European signal processing conference (EUSIPCO). IEEE, pp 91–95
- Valsaraj A, Madala I, Garg N et al (2021) Alzheimer’s dementia detection using acoustic & linguistic features and pre-trained BERT. In: 2021 8th International conference on soft computing & machine intelligence (ISCMI). IEEE, pp 171–175
- Vats NA, Yadavalli A, Gurugubelli K, et al (2021) Acoustic features, BERT model and their complementary nature for Alzheimer’s dementia detection. In: 2021 Thirteenth international conference on contemporary computing (IC3-2021), pp 267–272
- Vats NA, Barche P, Mirishkar GS, et al (2022) Exploring high spectro-temporal resolution for Alzheimer’s dementia detection. In: 2022 IEEE International conference on signal processing and communications (SPCOM). IEEE, pp 1–5
- Vetráb M, Egas-López JV, Balogh R et al (2022) Using spectral sequence-to-sequence autoencoders to assess mild cognitive impairment. In: ICASSP 2022—2022 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, pp 6467–6471
- Voleti R, Liss JM, Berisha V (2019) A review of automated speech and language features for assessment of cognitive and thought disorders. IEEE J Sel Top Signal Process 14(2):282–298
- Wen B, Wang N, Subbalakshmi K et al (2023) Revealing the roles of part-of-speech taggers in Alzheimer disease detection: scientific discovery using one-intervention causal explanation. JMIR Formative Res 7(1):e36590
- Woszczyk D, Hedlikova A, Akman A et al (2022) Data augmentation for dementia detection in spoken language. arXiv preprint [arXiv:2206.12879](https://arxiv.org/abs/2206.12879)

- Xue C, Karjadi C, Paschalidis IC et al (2021) Detection of dementia on voice recordings using deep learning: a Framingham heart study. *Alzheimer's Res Ther* 13:1–15
- Yadav V, Kumar R, Azad C (2022) A filter-based feature selection approach for the prediction of Alzheimer's diseases through audio classification. In: 2022 2nd International conference on advance computing and innovative technologies in engineering (ICACITE). IEEE, pp 1890–1894
- Yamada Y, Shinkawa K, Kobayashi M et al (2021) Combining multimodal behavioral data of gait, speech, and drawing for classification of Alzheimer's disease and mild cognitive impairment. *J Alzheimers Dis* 84(1):315–327
- Yamada Y, Shinkawa K, Nemoto M et al (2023) A mobile application using automatic speech analysis for classifying Alzheimer's disease and mild cognitive impairment. *Comput Speech Lang* 81:101514
- Yang Z, Dai Z, Yang Y et al (2019) Xlnet: Generalized autoregressive pretraining for language understanding. In: *Advances in neural information processing systems*, vol 32
- Yang L, Wei W, Li S et al (2022) Augmented adversarial self-supervised learning for early-stage Alzheimer's speech detection. *Proc Interspeech 2022*:541–545
- Ye Z, Hu S, Li J et al (2021) Development of the cuhk elderly speech recognition system for neurocognitive disorder detection using the dementiabank corpus. In: *ICASSP 2021—2021 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, pp 6433–6437
- Yuan J, Bian Y, Cai X et al (2020) Disfluencies and fine-tuning pre-trained language models for detection of Alzheimer's disease. In: *Interspeech*, pp 2162–6
- Zheng C, Bouazizi M, Ohtsuki T (2022) An evaluation on information composition in dementia detection based on speech. *IEEE Access* 10:92294–92306
- Zhu Y, Liang X, Batsis JA et al (2021a) Exploring deep transfer learning techniques for Alzheimer's dementia detection. *Front Computer Sci* 3:624683
- Zhu Y, Obyat A, Liang X et al (2021b) Wavbert: exploiting semantic and non-semantic speech using wav2vec and BERT for dementia detection. In: *Interspeech*. NIH Public Access, p 3790

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